

Vertical Shaft Sinking

15/05/2020

Starting of Shaft I

- *Conventional drilling, blasting method is not applicable at beginning of excavation.
- *First 50m (from Surface) is sunk before full fladged sinking by installing head gear & winder.

Drilling Blasting I

- *Conventional shaft sinking is done by drilling, blasting, mucking, lining.
- *This cycle is repeated through out sinking.
- *Good drilling, blasting should be followed.
- *Correct size & shape of excavation as planned.
- *Even surface of face (shaft bottom) makes mucking & drilling of next round.
- *Uniform size of broken rock for effective mucking operation.
- *Safe & economic operation.

Drilling Blasting II

- *In shaft sinking, water proof explosive of higher strength is used.
- *Electric as well as non electric delay detonator is used for blasting.
- *Normally, indirect initiation is followed in shaft sinking.
- *Holes of 2.5 to 3m are drilled in 5m dia shaft in hard rock.
- *Hole diameter is 28 to 32 mm.
- *Deeper hole reduces blasting efficiency.

Drilling Blasting III

- *Long hole consumes more drilling time, explosive for per meter shaft progress.
- *Increasing hole diameter, blasting & Drilling efficiency goes up.
- *Hole diameter more than 32mm requires deployment of Drilling Jumbo.
- *Longer hole consumes more drill time, explosive, for per meter shaft progress.
- *In case of too long hole, blasting efficiency diminishes in upper part of hole, crushing of rock is not sufficient.

Drilling Blasting IV

- *It is because longer part of hole is free from explosive.
- *Deeper part, rock material is crushed.
- *Benefit of using large diameter , longer hole reaches at optimum value (level). Further increasing hole diameter & length does not give benefit for said diameter shaft.
- *“Hole loading factor” ratio of length of hole charged with explosive by total length of hole.

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Drilling Blasting V

*Hole diameter 32 to 36mm, Loading factor 0.6 -0.7

*Hole diameter 45 to 50mm, Loading factor 0.45 -0.5

Hole Length I

- *Hole length is different from hole depth.
- *Hole length is shortest distance between free face & end of hole.
- *Hole inclination create this difference.
- *Length of hole being determined by geology & mining conditions.
- *Other factors for deciding length of holes are drilling equipment, explosive material & charging, organization of shaft sinking.
- *Hole diameter depends on rock type, water inflow in hole.

Hole Length II

- *Shaft sinking consist of sequence of drilling, blasting, mucking, hoisting, lining.
- *Shorter holes are chosen when one cycle per shift is planned.
- *Most desirable hole length will be length that assures that unit advance with minimum time & cost achieved.
- *An optimum hole length should be found from minimum cost principle.

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Hole Length III

- *Main cost contributors in shaft sinking are drilling of holes, explosive & mucking.
- *Depth of round is a contributory factor in total cost of sinking.
- *Hole deviation influence performance of drilling & blasting.
- *Hole 2 to 2.5m long deviates 2-3%
- *Hole 4 to 5m long deviates 5-10%

Hole Length IV

- *Normally for shaft sinking, hole length is 3m.
- *There is a trend to increase hole length for faster shaft sinking.
- *Required hole length (drilling pattern) has to be drawn at designing state it self.
- *Provision to change drilling pattern should be kept based on ground condition.

Drilling Round I

- *Drilling pattern has strong influence on sinking performance.
- *Drilling pattern should ensure following.
- *Accuracy in shape of excavation.
- *Clean separation of rock from bottom of excavation (shaft face).
- *Even & sufficient comminuting of rock.
- *Avoiding cut off of neighboring active hole.
- *Avoiding damage to near by shaft installation.

Drilling Round II

- *Drilling pattern depends on shape of shaft section, rock strength, cleavage, dip of strata, water inflow, hole loading structure.
- *In circular shaft with small diameter & moderate strata dip, holes are placed on concentric circles, down centre of shaft.
- *Number of circles are 3 to 5 depending up on shaft diameter.
- *Central hole is vertical & shorted about $\frac{2}{3}$ of other holes.

Drilling Round III

- *These holes inside conical excavation by cut holes, specially recommended in strong rock.
- *Holes of first (central) circle, wedge hole inclined to wards shaft centre.
- *This holes are first to open & additional free surface made.
- *This free face makes easier & more efficient work for remaining external holes.
- *Holes in outer most circle is drilled out ward inclination of 65 to 75 degree.
- *4 to 10 holes are drilled for first circle.

Drilling Round IV

- *Harder rock needs more first cut holes .
- *For explosive cartridge 32mm dia, & shaft dia is D , following are drilling circle diameter.
- *With 3 circle $0.37D$, $0.66D$ $0.93D$
- *With 4 circle $0.35D$, $0.54D$ $0.7D$, $0.93D$
- *With 5 circle $0.27D$, $0.43D$ $0.6D$, $0.73D$. $0.93D$
- *Ratio of hole number
- *3 circles 1-2-3
- *4 circles 1-2-3-4
- *5 circles 1-2-3-4-5

Drilling Round V

- *For explosive cartridge 45mm dia, & shaft dia is D , following are drilling circle diameter.
- *With 3 circle $0.3D$, $0.6D$ $0.95D$
- *With 4 circle $0.25D$, $0.48D$ $0.72D$, $0.96D$
- *Ratio of hole number
- *3 circles 1-3-6
- *4 circles 1-2-3-5
- *Inclination of holes varies from 65 to 85 degree.
- *Holes farther from shaft centre, more vertical they are.

Drilling Round VI

- *Distance between adjustment holes on a circle should not exceed 0.8 to 0.9 for 32 & 36mm cartridge.
- *For 45mm cartridge , it should be in range of 1 to 1.2 m.
- *Last circle of rib holes are to leave shaft wall smooth.
- *Last circle holes are drilled to wards shaft wall 65 to 85 degree.
- *Inclination of hole should be less for weak formation & more for stronger formation.
- *In very weak rock they can be slightly inclined towards center of shaft.

Drilling Round VII

- *Rib holes are placed in circle with a diameter of 0.6m less than shaft diameter, when coherence factor of rock is 2 to 8.
- *This distance is 0.3m when coherence factor of rock is 9 to 20.
- *Distance from shaft wall has to be such that holes should not project from designed shaft section.
- *For rock with coherence factor more than 8, hole should protrude about 100mm to 200mm out side shaft contour.

Drilling Round VIII

- *Distance between rib holes in moderate strength rock 0.9m to 1.2m & in strong rocks 0.7m to 0.9m.
- *Some time it is required to place holes as close as 0.2 to 0.3m.
- * When strata have a dip of more than 45degree, wedge holes are located along strike, increasing utilization of holes along stratification.
- *Over breakage of rock will be large on side of HW strata.

Drilling Round IX

- *Distance of rib holes from shaft contour on dip side is 0.25 to 0.4m in moderate strength strata & 0.15 to 0.25 m in strong rocks.
- *In incline strata rock being blasted tends to form pile on one side of shaft.
- *Piling of muck on one side creates difficulty in placing bucket, thus slowing down mucking progress.
- *In rectangular shaft, blasting holes are distributed similarly to horizontal drift.

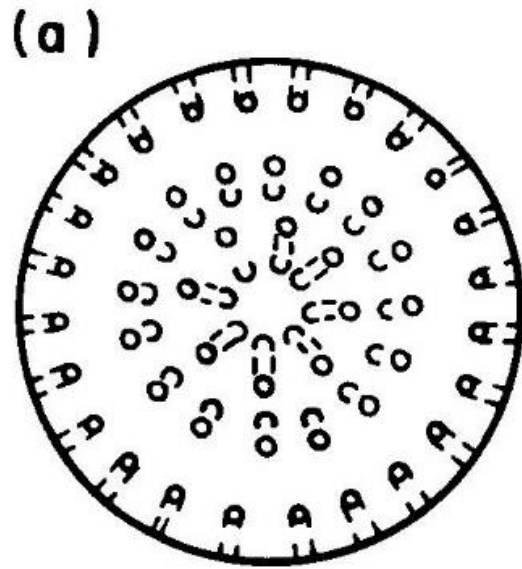
Drilling Round X

- *Typical & effective system of drilling is wedge & pyramid.
- *Step pattern is used for conditions of large water inflow (80m³/hour)
- *Deeper part works as a sump for pump.
- *Large diameter hole is drilled parallel to axis of shaft by drill jumbo.
- *This hole provide free face for wedge.
- *This method increases efficiency of sinking & reduces consumption if explosive.

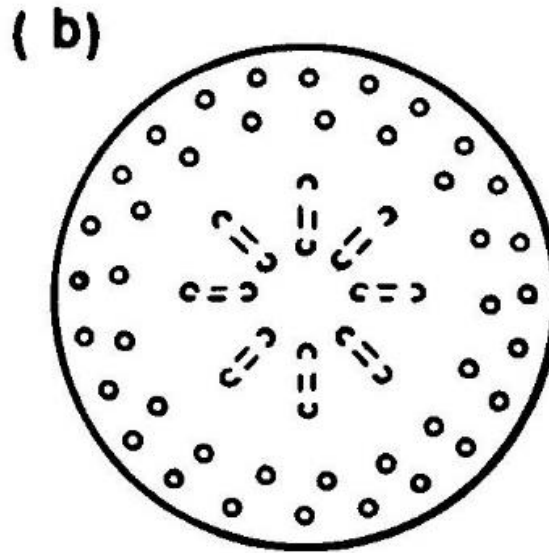
Drilling Round XI

- *Blasting includes indirect initiation (back) .
- *Electric milli second detonator is used for blasting.
- *Same delay is used for a particular circle.
- *Cartridge with detonator goes first.
- *A centrally located pile of muck implies a correct blast.
- *Muck piles on one side of shaft indicates of misfire or non uniform hole charging.
- *When misfire has been located , an additional shot hole is drilled parallel to shot out remnant stamp of rock & blasting material in unfired hole.

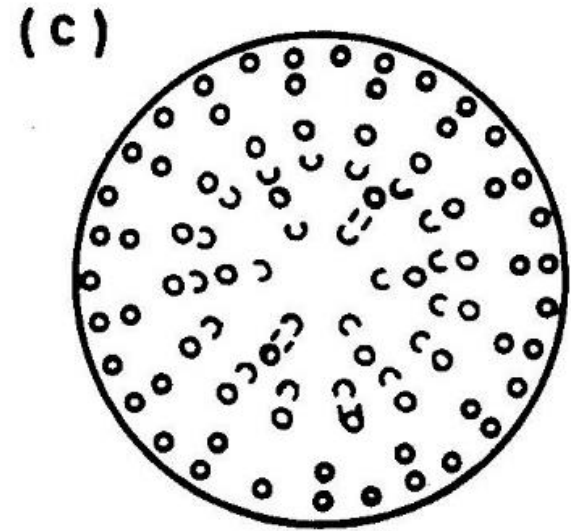
Drilling Pattern in Circular Shaft



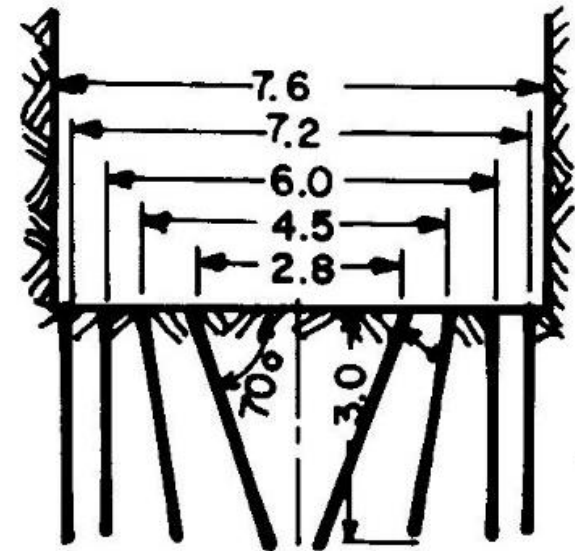
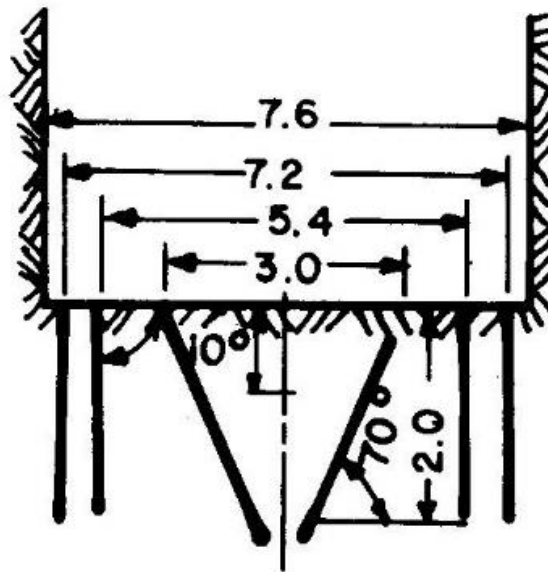
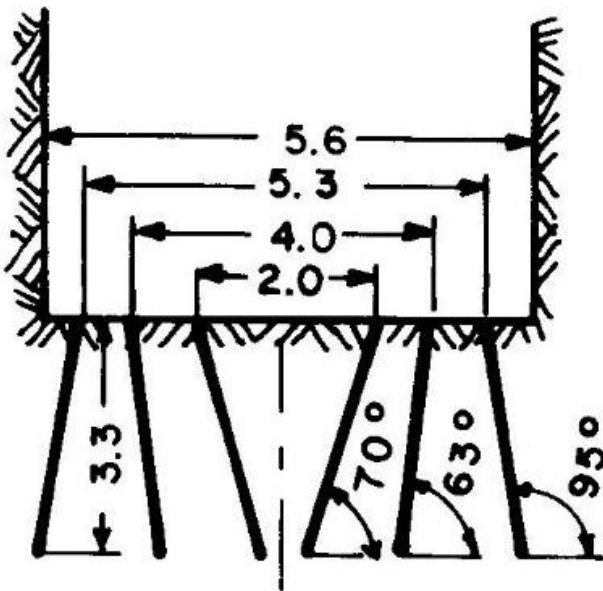
Siltstone



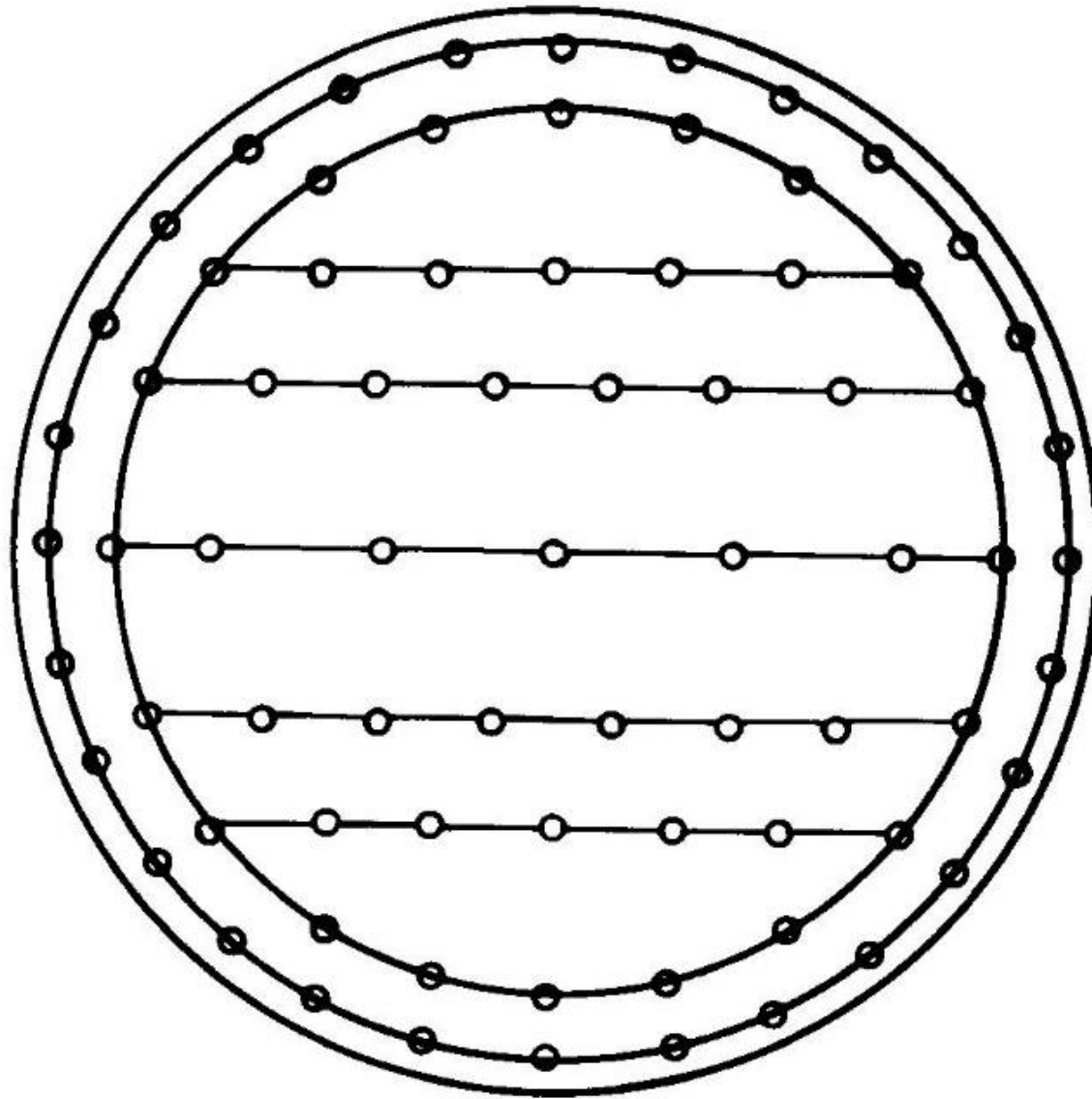
Shale



Sandstone



Drilling Pattern I

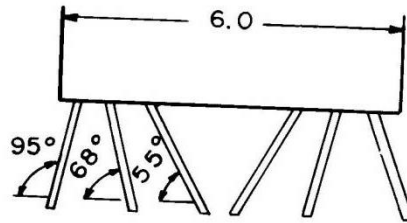
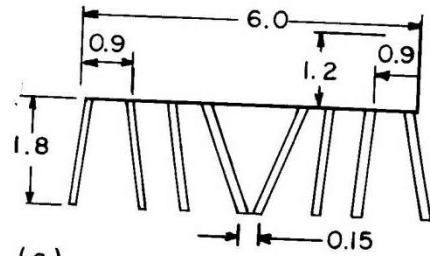
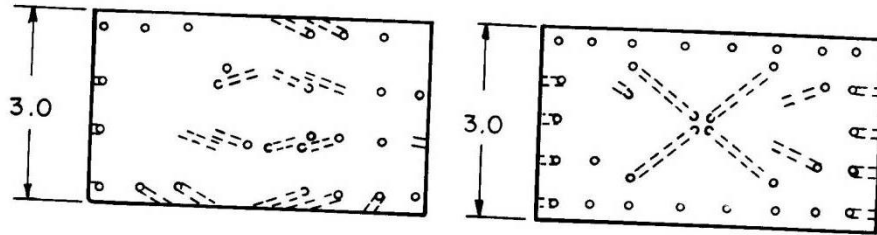


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Fig. 17.4.24. Example of drillhole distribution in shaft sunk in inclined strata.

Drilling Round I

*Drilling round in rectangular shaft.

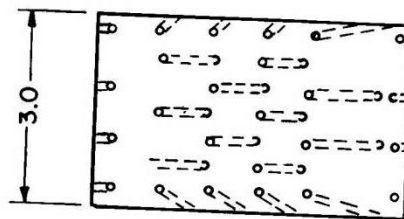
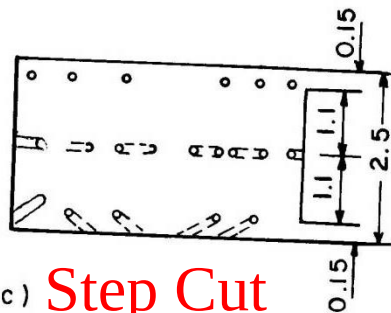
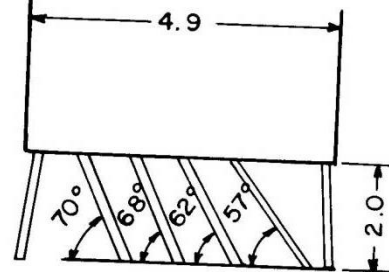
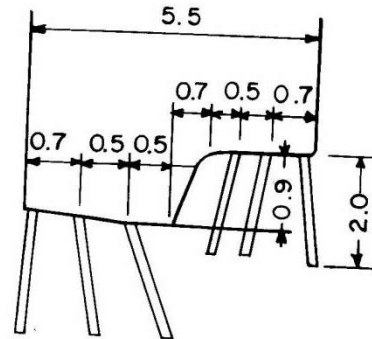


(a)

(b)

Wedge Cut

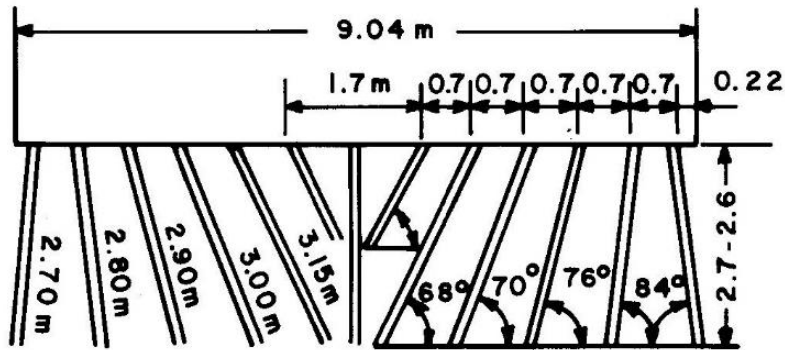
Pyramid Cut



(c)

Wedge Side Cut

Drilling Round Circular Shaft



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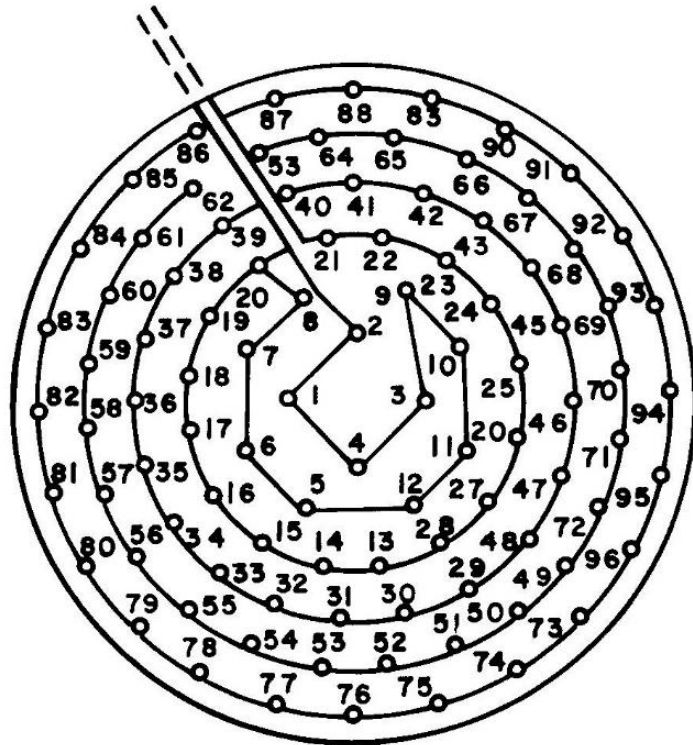


Fig. 17.4.26. Drilling pattern for shaft 27.5 ft (9.04 m) in diameter.
Conversion factor: 1 ft = 0.3048 m.

Shaft Drilling Jumbo I

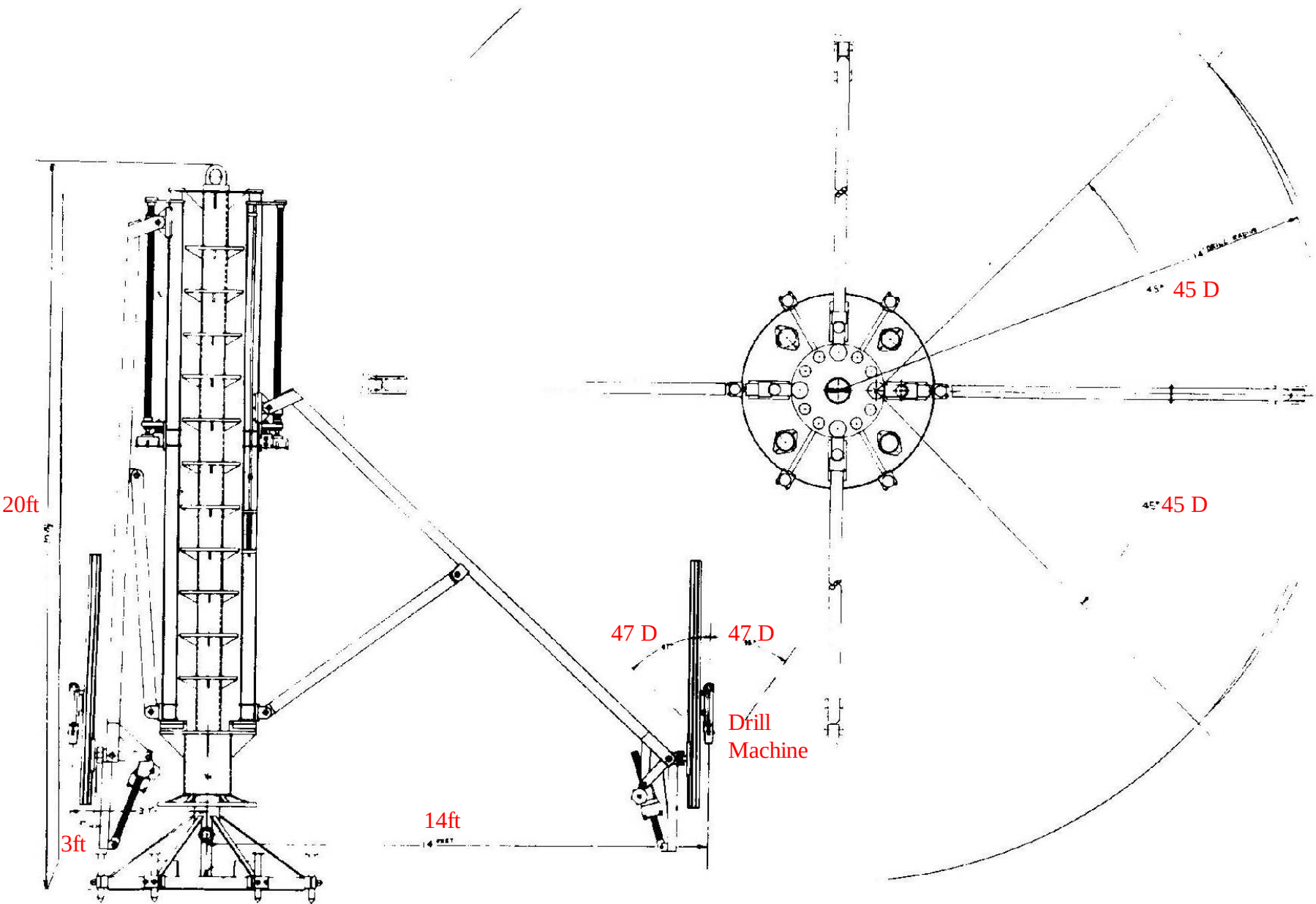


Fig. 17.4.27. ASJB-3B Shaft Jumbo (courtesy: Acme Machinery Co.). Conversion factor: 1 ft = 0.3045 m.

Blasting I

- *After drilling of holes, holes are cleaned from drill cutting by compressed air.
- *Explosive & detonator are taken in shaft face for blasting.
- *Priming of explosive done at face, shaft bottom.
- *Precaution has to be taken while lowering explosive & cartridge.
- *Explosive & detonator should not be subjected any type of jerk, pressure.
- *Blaster first charges primary cartridge in hole (indirect initiation)

Blasting II

- *Holes are charged with explosive, then plugged with stemming material.
- *Wedge holes are charged with 0 delay detonator.
- *Subsequent circles shall have increasing delay.
- *Detonator & blasting wire should be well insulated.
- *Insulation should also hazard from stray current.
- *After charging holes, end of detonator lead cord are short circuited, until they are connected to blasting cable.
- *Clay & sand mixture is used as stemming material.

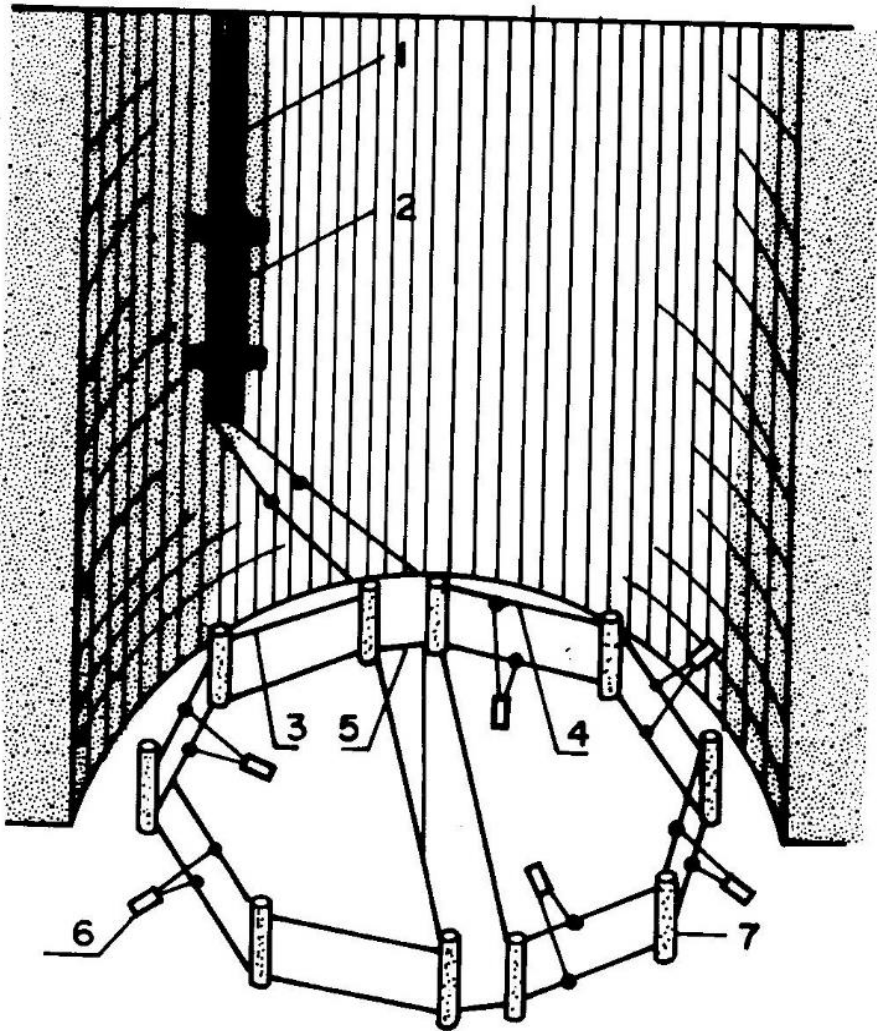
Blasting III

- *In wet shaft, flooding water acts as stemming material.
- *Loading time for each hole is 6 to 8 minutes.
- *Whole round charging takes 60 to 90 minutes.
- *Series connection of detonator is avoided due to low reliability.
- *Failure of one detonator should cause misfire of all holes.
- *A mixed connection is done so that group of holes are connected in series.

Blasting IV

- *Groups are connected parallel to main blasting cable.
- *After all connections are done, face light & other shaft fittings are pulled up to avoid damage.
- *Firing of hole is done from surface or from top of shaft.

Blasting Round I



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Fig. 17.4.28. Three-ring connection of electric caps.

- 1. shot cable
- 2. protecting pipe
- 3-4. active antenna
- 5. passive antenna
- 6. electric caps (in holes)
- 7. wooden stick supporting antennas.

Ventilation I

- *Shaft is ventilated by auxiliary ventilation fan located at surface of top of sinking shaft near shaft collar.
- *Metal ventilation duct is fixed at shaft wall for delivering air at shaft bottom.
- *Ventilation duct is extended along with progress of shaft sinking.
- *In case of deep shaft, in line booster auxiliary fan is installed to sustain fan pressure.

Mucking I

- *Mucking in shaft is most time & effort consuming operation.
- *About 50 to 60% cycle time is engaged in mucking.
- *Loader cactus grab are used for loading muck to bucket.
- *Mucking efficiency is dependent on degree of rock fragmentation & height of bucket.
- *For mechanized mucking, medium size fragmentation, approximately 125mm is most efficient.

Mucking II

- *Generally, in muck pile after blast, 70 to 80% of material is fine fragmentation & rest is composed of larger fragmentation.
- *Mucking efficiency goes down along with depth of shaft.
- *Type of mucking machines depends on shaft shape, size, depth, location.
- *Scraper is used in rectangular shaft or very large diameter shaft (under ground bunker).
- *Over head loader with crawler chain.

Mucking III

- *Cryderman mucker.
- *Cactus grab light.
- *Cactus grab heavy, on central pivot.
- *Some mucking system are driven by operator.
- *Hand guided grab at face when hanging from platform.
- *Attached to shaft wall (cryder man) & operated from control cabin.
- *Heavy cactus grab unit have a rotating arm on a central pivot below stage.

Mucking IV

- *Large mucker can be of 1m³ capacity
- *Generally, large capacity muckers are used for deep shaft with large diameter.
- *Mucking capacity is expressed by
$$P = 3600 k V/t$$
 - P > mucking capacity m³/hr
 - k > fill factor, 0.8 to 0.9 for sand stone, 1.1 to 1.2 for shale.
 - V > bucket volume in m³
 - t > mucking time in sec

Mucking V

*Actual efficiency is found by calculating actual mucking time & all-time losses for unproductive work such as mucker time to work position, time of temporary support when mucking has to be suspended & time for final cleaning.

Sinking Operation I

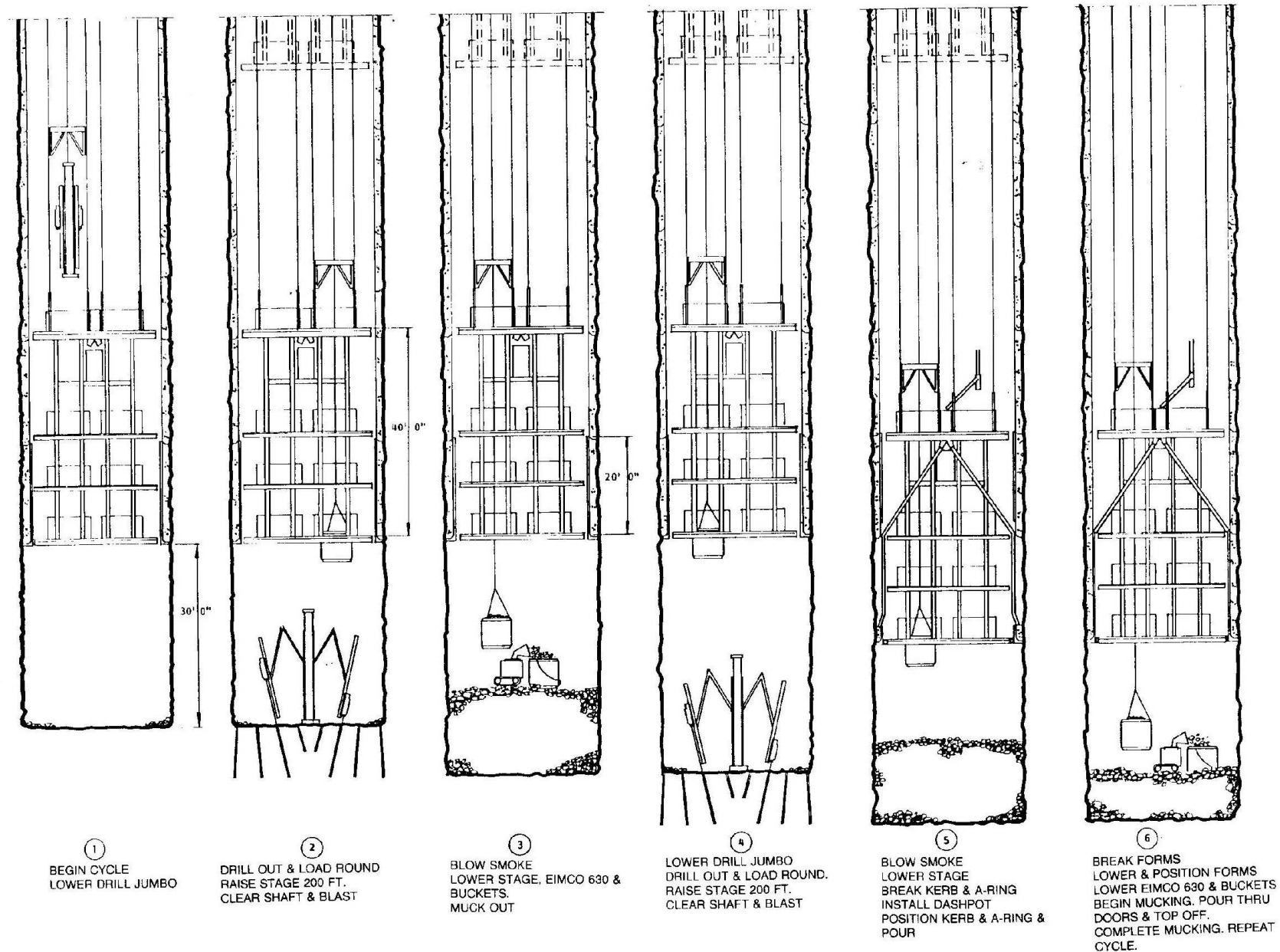
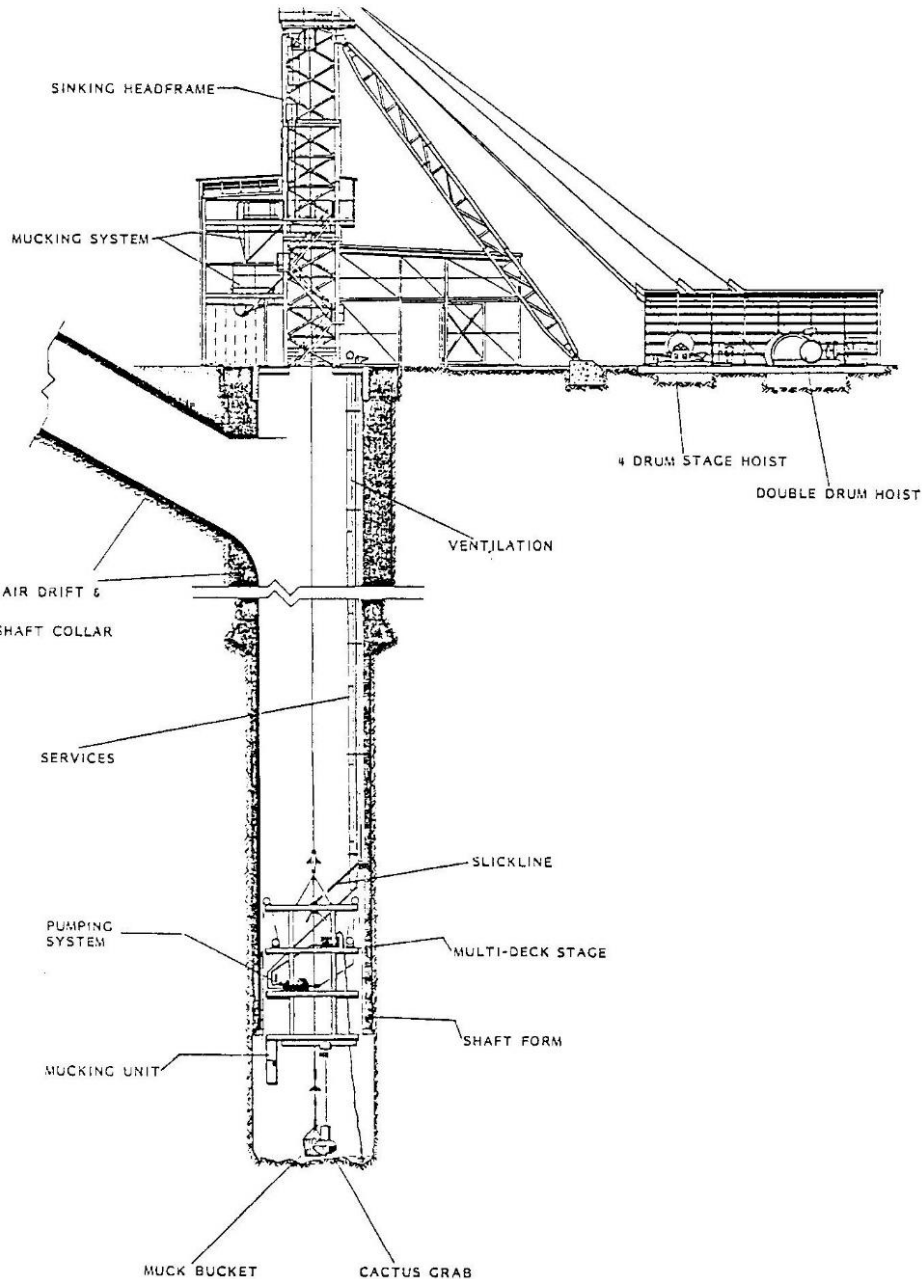


Fig. 17.4.29. Schematic of sinking operations. (After Hynd, 1982.) Conversion factor: 1 ft = 0.3048 m.

Drilling Round I

CONSTRUCTION OF DEVELOPMENT OF

*Shaft Sinking Arrangement



Dewatering I

- *Most shaft intersect one or more water course.
- *Shaft dewatering is an essential part of sinking.
- *Characteristic of shaft dewatering are changing water inflow, changing pump head, water containing rock & muck particle, periodic change in pump system.
- *Dewatering system can be considered for limited water inflow or substantial water inflow in shaft.
- *Water into shaft can be diminished by initial drainage of rock mass, sealing if water courses, tapping water.

Dewatering with Bucket II

- *Removal of water with muck being hoisted is simplest dewatering system.
- *A small portable pump is usually driven by compressed air with head 18m & out put 0.05 to 0.5 m³/hour.
- *It could be any other type of electric pump approved for shaft face pumping.
- *Usually kibble (bucket, skip) exclusively filled with water is periodically hoisted up, depending on need.
- *Limit of bucket dewatering system derived from condition not to flood face shooting & ventilation.

Dewatering with Bucket II

$$q > 0.8 D_{sq} (k + 0.1) / t$$

q = water inflow in m_{cub}/h

t = pumping type considering blasting work

k = coefficient characterizing size of void in muck

(0.5 to 0.6 for hard rock & 0.3 to 0.4 for weak rock)

D = bucket diameter

l = depth of blast hole

0.1m is allowable water level in shaft.

Dewatering III

*Experience shows that limit for this system of dewatering for 150 m deep shaft is 3m³/h.

Dewatering pumping I

Face pump

- *Face pump is generally compressed air driven membrane pump, which can handle dirty water accumulated at shaft bottom.

Hanging pump

- *Hanging pump are used to control large inflow, when required.
- *Head of this pump is 200m to 400m.
- *This pump can be easily lifted up during blasting

Dewatering pumping II

Dewatering with air lift

- *Face pump is generally compressed air driven membrane pump, which can handle dirty water accumulated at shaft bottom.

Hanging pump

- *Hanging pump are used to control large inflow, when required.
- *Head of this pump is 200m to 400m.
- *This pump can be easily lifted up during blasting

Dewatering II

Dewatering with air lift

- *Air lift pumps are powered by compressed air.
- *There are two main system (a) parallel, where air pipe runs beside water pipe (b) concentric in which air pipe is inside water pipe.
- *Air lift require suitable submersion, they are applicable in dewatering of flooded shafts.
- *They can handle dirty water even quick sand.

Dewatering Intermediate Pumping I

- *When sinking a deep shaft, intermediate pumping has to be applied.
- *Number of stage depends on depth.
- *On level chosen as stage, overflow chambers are built.
- *Pumping stage can be arranged as described below.
- *Face pump delivering water to overflow chambers & from then it is pumped by stationary pumps to surface or another chamber.

Dewatering Intermediate Pumping II

- *Hanging pump delivering water to over flow chamber from where stationary pumps send it upward.
- *Face pump pumps water to flow chamber where an air lift delivers it up to surface.

Lining I

- *Factor determining type of lining are hydrogeology, condition, shaft functions, planned life time, shape of shaft section, depth of shaft, availability of construction material, construction cost, geotechnical properties of rock, hydrologic condition. Chemical activity (corrosiveness) of water.
- *Modern shaft often have automatically operated hoisting gear (moister sensitive), they should be dry.
- *Main shaft is usually planned for entire mine life.
- *They should be constructed to minimize repair & maintenance time.

Temporary Lining I

- *Temporary support is placed to protect crew & equipment against falling rocks from exposed shaft wall before permanent lining is placed.
- *Temporary support is used when face is advancing, leaving a longer unlined shaft wall 6 to 40m.
- *Work at face is then suspended & temporary support is dismantled allowing positioning of concrete form.
- *When removing of temporary lining creates safety problem, it can be left & concrete poured over it.

Temporary Support System I

- *Steel ring in profile are assembled & suspended from round with help of 25mm steel hanger.
- *Each ring consist of six to eight separate segments bolted together.
- *There is provision to change diameter to fit excavation.
- *Board, iron grate or sheet are placed behind ring.
- *Final ring is supported by pin (in to shaft wall) around shaft perimeter.
- *Vertically at every 4th or 6th ring are supported.

Temporary Support System II

*Wooden wedges are driven to make ring tight.

Permanent lining Timber I

- *Timber lining is rarely used for important & main shaft.
- *It is only used for auxiliary shaft with short life span (inter level blind shaft).
- *Timber lining can be full with sets lying one on top of another in more difficult ground conditions.
- *In better condition, on post with distance of particular set 0.5 to 1.5m.
- *In both arrangement, every 8 to 15 m, bearing beams are wedged to get entire structure tightened into surrounding rock.

Permanent Lining Brick I

- *Brick lining is no more practiced.
- *It is difficult to do brick lining in a mechanized shaft sinking.
- *Brick lining is simple & easy to construct.
- *Brick lining can take load from beginning itself (immediately).
- *Brick lining is easy to repair & resist to corrosion.
- *Brick lining suffers from following disadvantages.
- *Time & labour consuming erection.
- *Low strength.

Permanent Lining Brick II

- *High cost (where labour is expensive).
- *Substantial permeability.

Permanent Concrete Block Lining I

- *Concrete block lining is improved brick lining process.
- *There is decreased number of seam & higher strength of concrete shaft wall.
- *This lining is less labour intensive when compared to brick lining.

Monolithic Concrete Lining I

- *It is most popular shaft lining.
- *It has possibility of complete mechanization of construction process with slip or switch form & by transporting concrete through slick line.
- *Good bond between lining & shaft wall
- *Shaft footing (formation) is not required.
- *Decrease in labour intensity by 30 to 40% in compared with brick lining.
- *High strength, resulting in less required excavation.
- *

Monolithic Concrete Lining II

- *Moderate strength concrete (17 to 25 Mpa) is sufficient for lining purpose.
- *Concrete is made of natural aggregate, Portland or slag cement.
- *Concrete mixture is prepared in vicinity of shaft in a mechanized concrete plant.

Monolithic Concrete Lining III

- *Concrete is send down shaft by one or two pipe line
- *Concrete can also be transported by hopper.
- *Minimum lining thickness is 200mm & maximum is 800mm.
- *While sinking in series, short segment of shaft (4-5m) in length are lined with application of collapsible steel form.
- *Form is relocated after every concrete set.
- *Lining made in this way is not a monolithic but consist of segment with seam.

1-C
Part-2
Vertical Shaft Sinking

15/05/2020

Monolithic Concrete Lining IV

- *When shaft is sunk in long lifts, a sliding concrete form can be applied.
- *This results in longer section of monolithic shaft lining.

Disadvantages

- *Less resistant to corrosive water.
- *Sensitivity to rock movement.
- *Inability to take load immediately after placement.
- *Difficulty of repair.

Shotcrete Lining I

- * Shotcrete lining can be made in following variants.
- * Regular shotcrete.
- * Shotcrete with rock bolts.
- * Reinforced shotcrete with bolts.
- * Shotcrete lining is applicable in dry rock shaft with good strength.
- * Especially in blind shaft of smaller diameter, where use of concrete is limited, shotcrete is very use full.
- * Shotcrete has very good binding with rock.
- * Shotcrete lining is strong & very tight.

Shotcrete Lining II

- *Shortcrete lining is strong & very tight due to low water / cement ratio (w/c).
- *Shortcrete is more often used in combination with rock bolt & mesh.
- *Prior to shortcrete, rock bolt & wire mesh are installed as temporary support.
- * Shortcrete covers mesh pinned to shaft wall with bolts & creates a strong but thin shell lining.
- *Shortcrete has very good binding with rock.

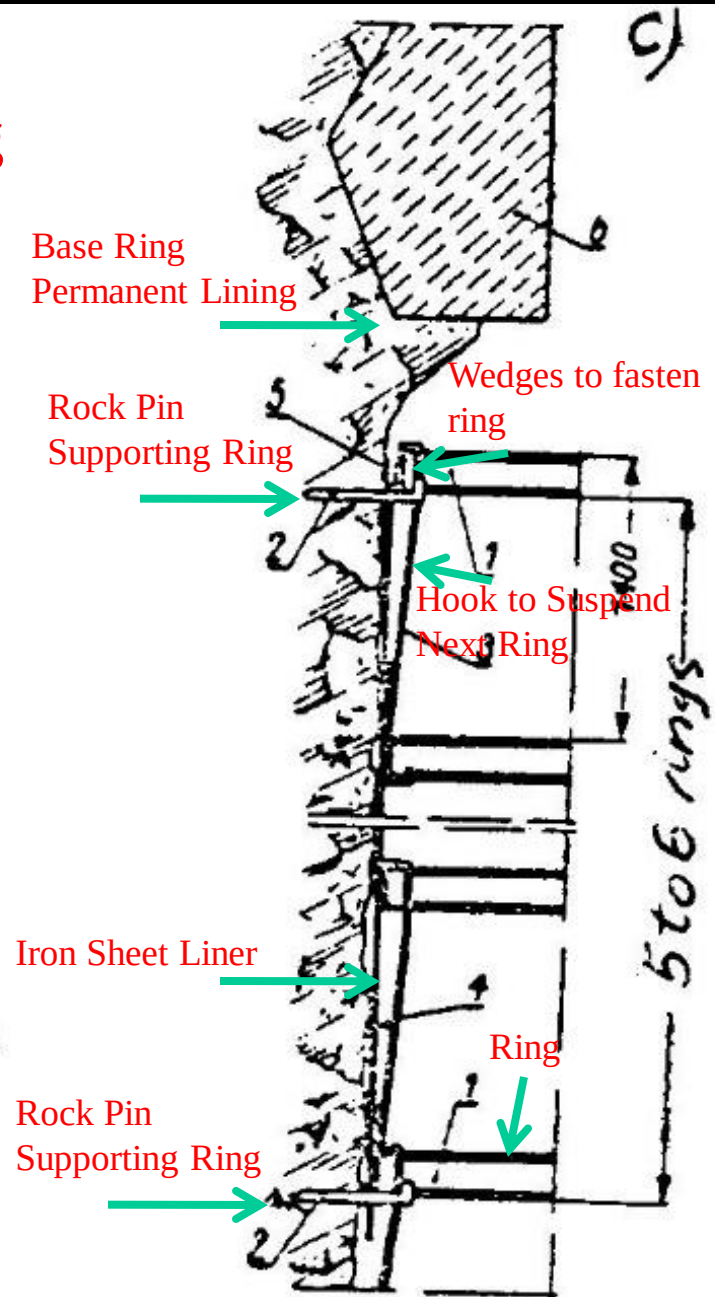
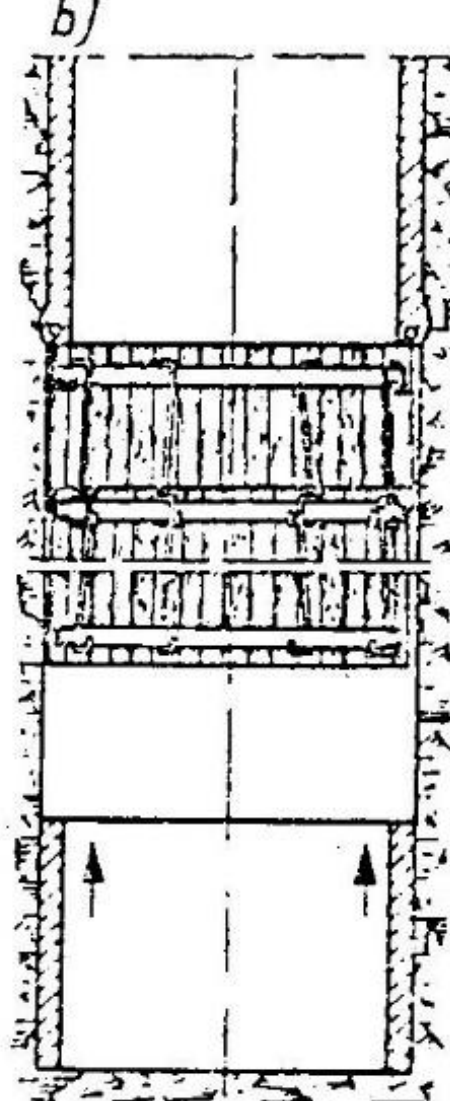
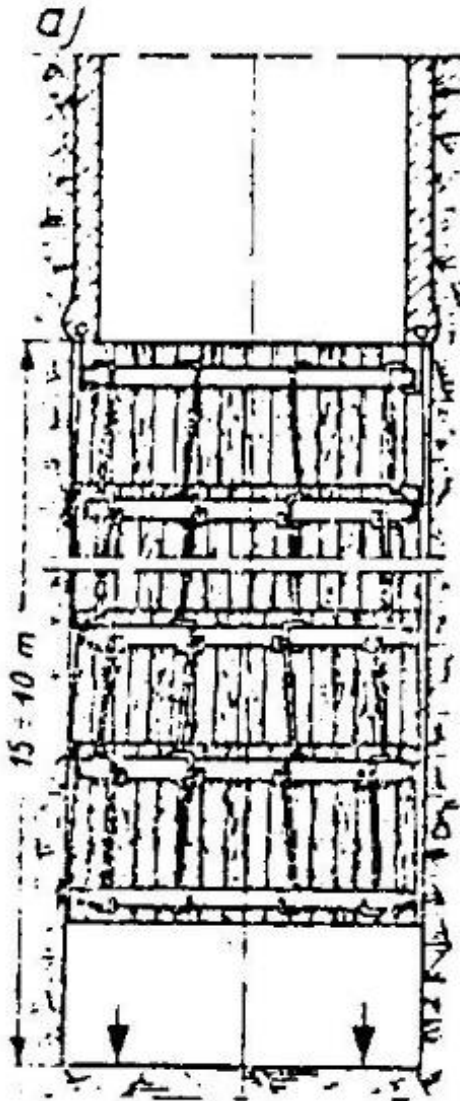
Rock Bolt Lining I

- *Rock bolt is used primarily in salt mine shaft.
- *Salt has creep characteristics to flow towards opening.
- *This shall exert an excessive pressure on a rigid shaft lining.
- *Pressure of lamination & over all tendency to weathering make condition worse.
- *Rock bolts are made of nonmetallic (plastic, which does not corrode) are used.
- *Some support system is used in inset chamber & other capital opening.

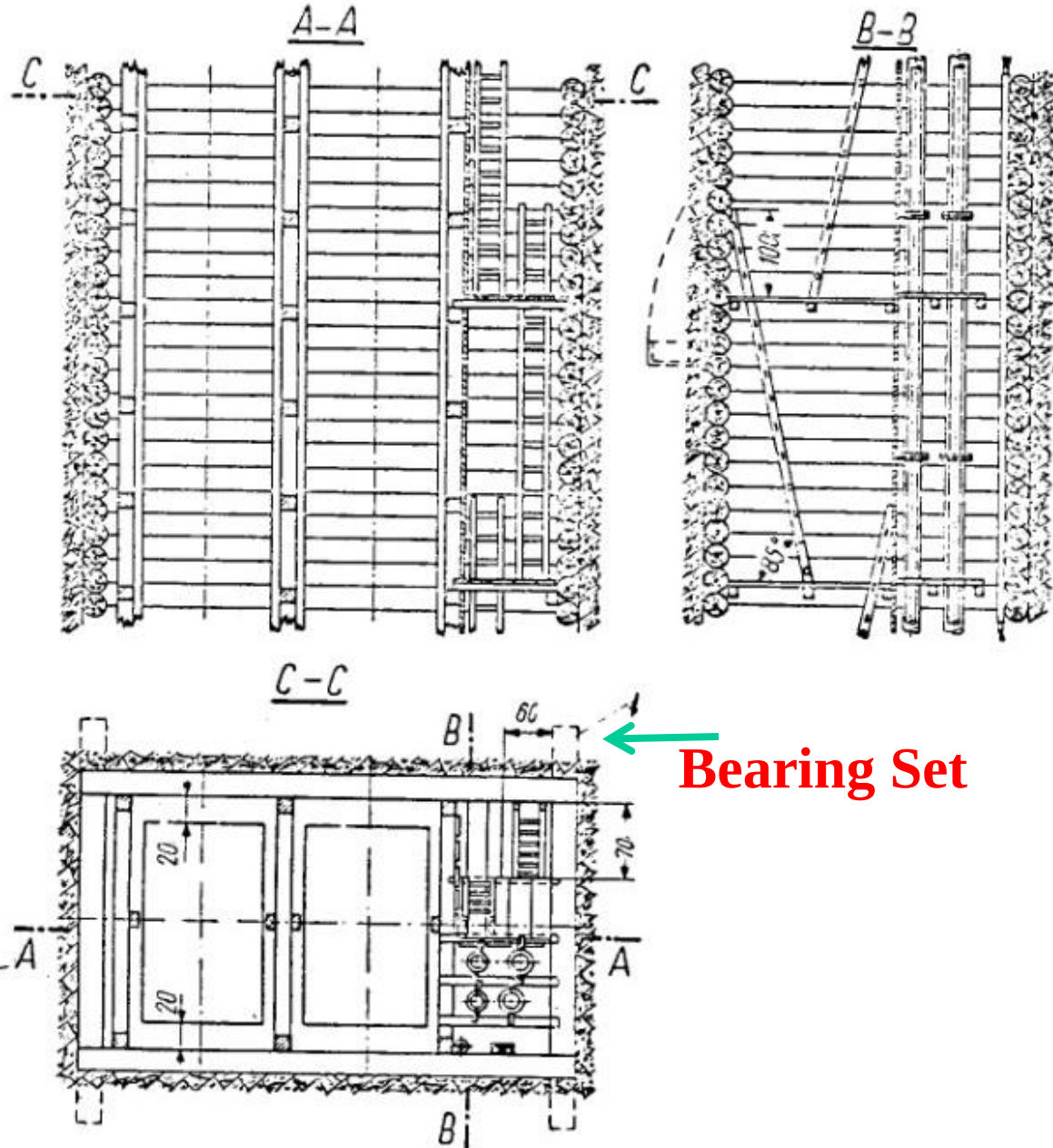
Drilling Round I

Sinking

Permanent Lining



Timber Permanent Lining

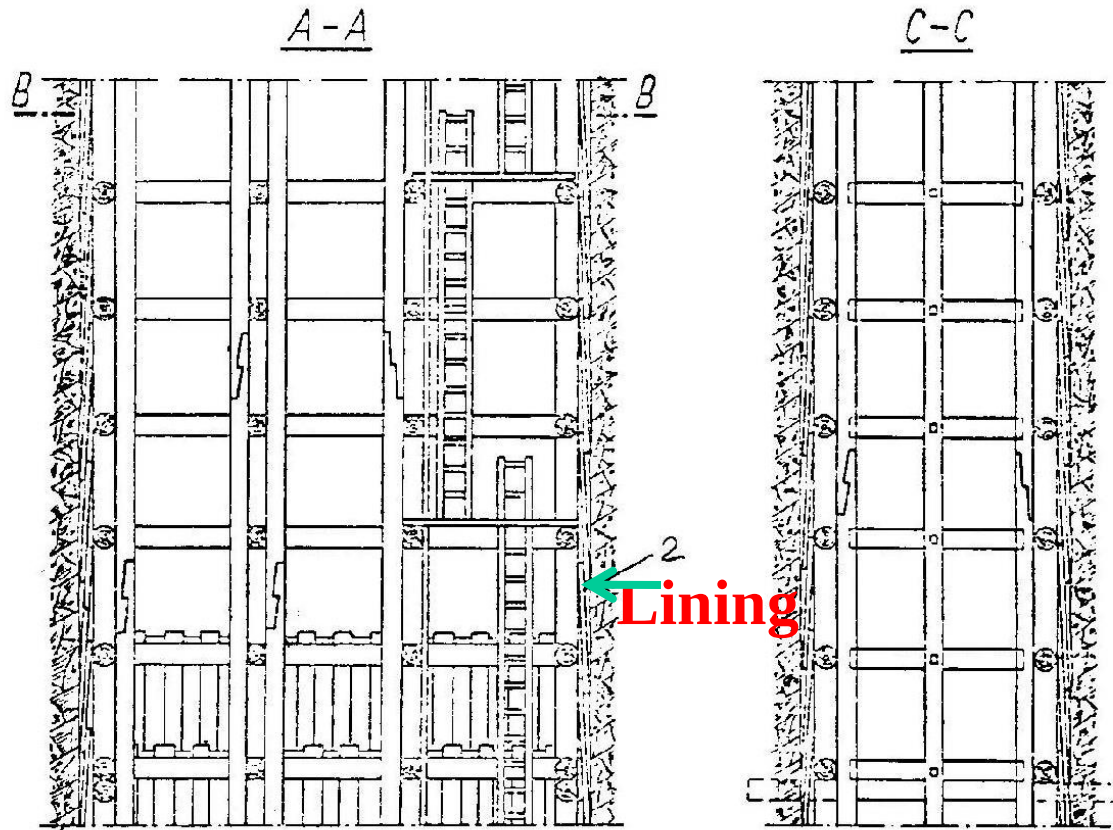


*Bearing set
are round 16-
22 cm dia or
square 14-20

Bearing Set

cm timber
used for sets.
Timber
should be
treated to
avoid fast
decaying.

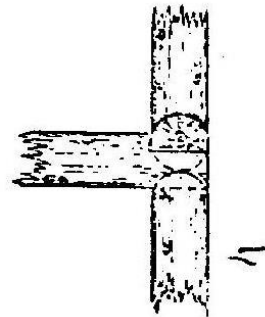
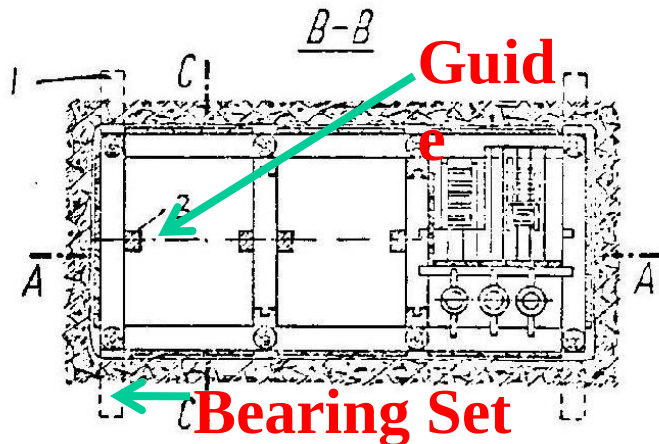
Timber Lining Full



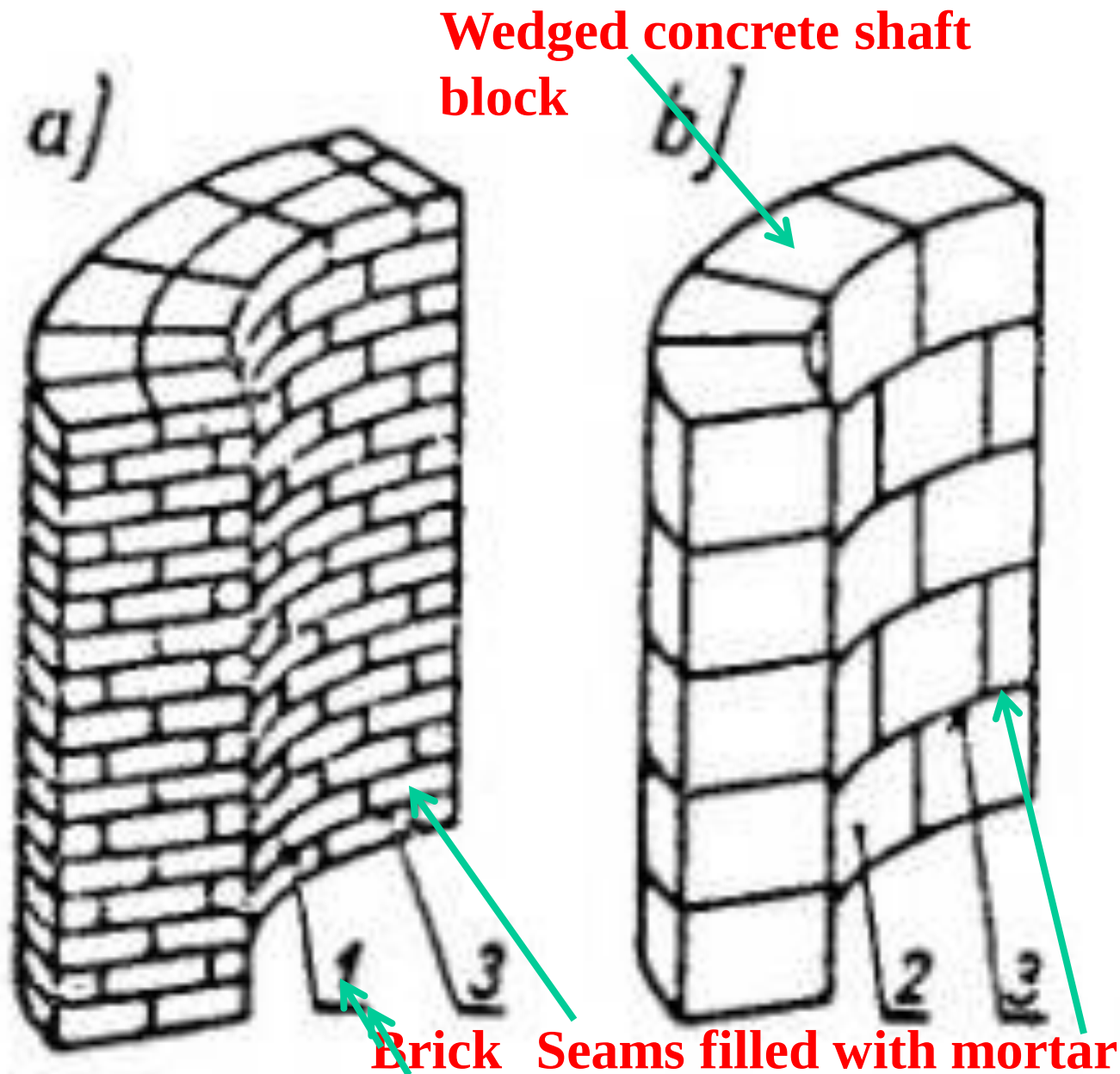
1- Bearing Set

2- Lining with 1 inch (25mm) boards.

3- Guide



Brick & Concrete Lining



*Permanent brick lining. Concrete block lining.

Concrete Bar Lining

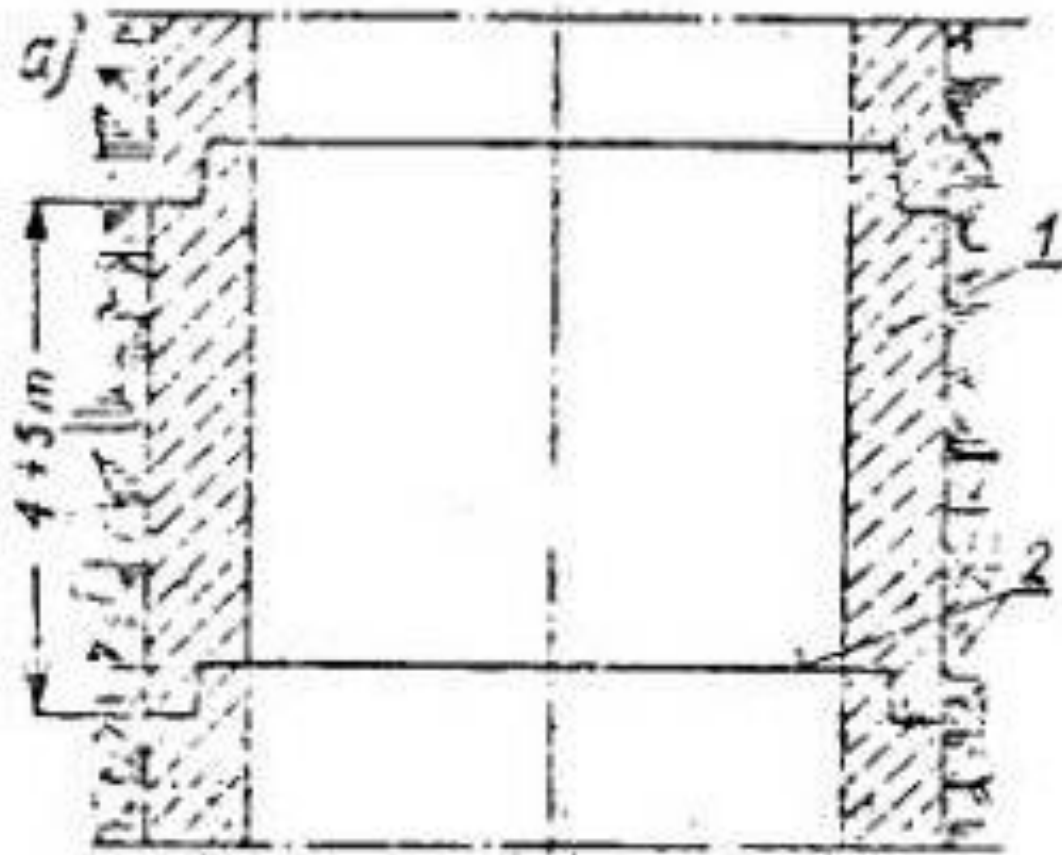


Fig. 17.4.35. Section through the shaft with concrete lining, made with application jump form.

- 1. concrete
- 2. seam.

*Concrete lining with pre casted beams

Reference I

*SME mining Engineering Handbook, 2nd Edition,
Page 1598 to 1607

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Reference Others

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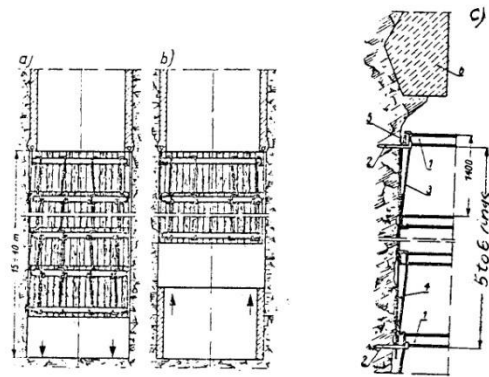


Fig. 17.4.31. Schematic of shaft sinking with long lifts and temporary support.

- a) phase of sinking
- b) phase of lining (permanent)
- c) temporary lining
1. ring
2. rock pin supporting the ring
3. hook to suspend next ring
4. iron sheet liners
5. wedges to fasten lining
6. base ring or permanent lining

Conversion factor: 1 ft = 0.3048 m.

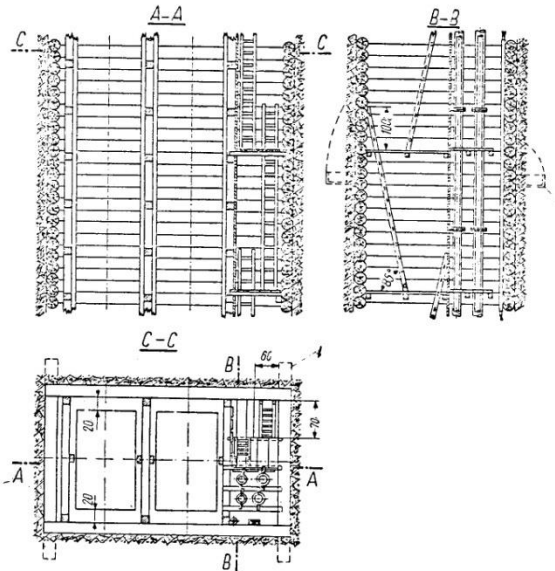


Fig. 17.4.32. Timber lining—full.

1. bearing set
- Round 16–22 cm dia. or square 14–20 cm timber used for sets. Timber should be treated to avoid fast decaying. Dimensions in cm. Conversion factor: 1 in. = 2.54 cm.

from synthetic materials (e.g., plastic, which does not corrode) are used. This same support system in salt is applied in insets, chambers, and other capital openings.

17.4.2.3 Specialized Methods of Shaft Sinking

CLASSIFICATION. Special methods of shaft sinking are used when the following conditions are encountered:

1. Incompetent, water-bearing rocks (e.g., quicksand).
2. Weak, unstable, soil-type rocks.
3. Competent fractured rocks, with high water inflows above 130 gpm (0.5 m³/min).

There are numerous special methods varying among themselves in applicability because of environmental conditions and range of depths. The characteristics of these are given in Table 17.4.9.

Sinking Operation I

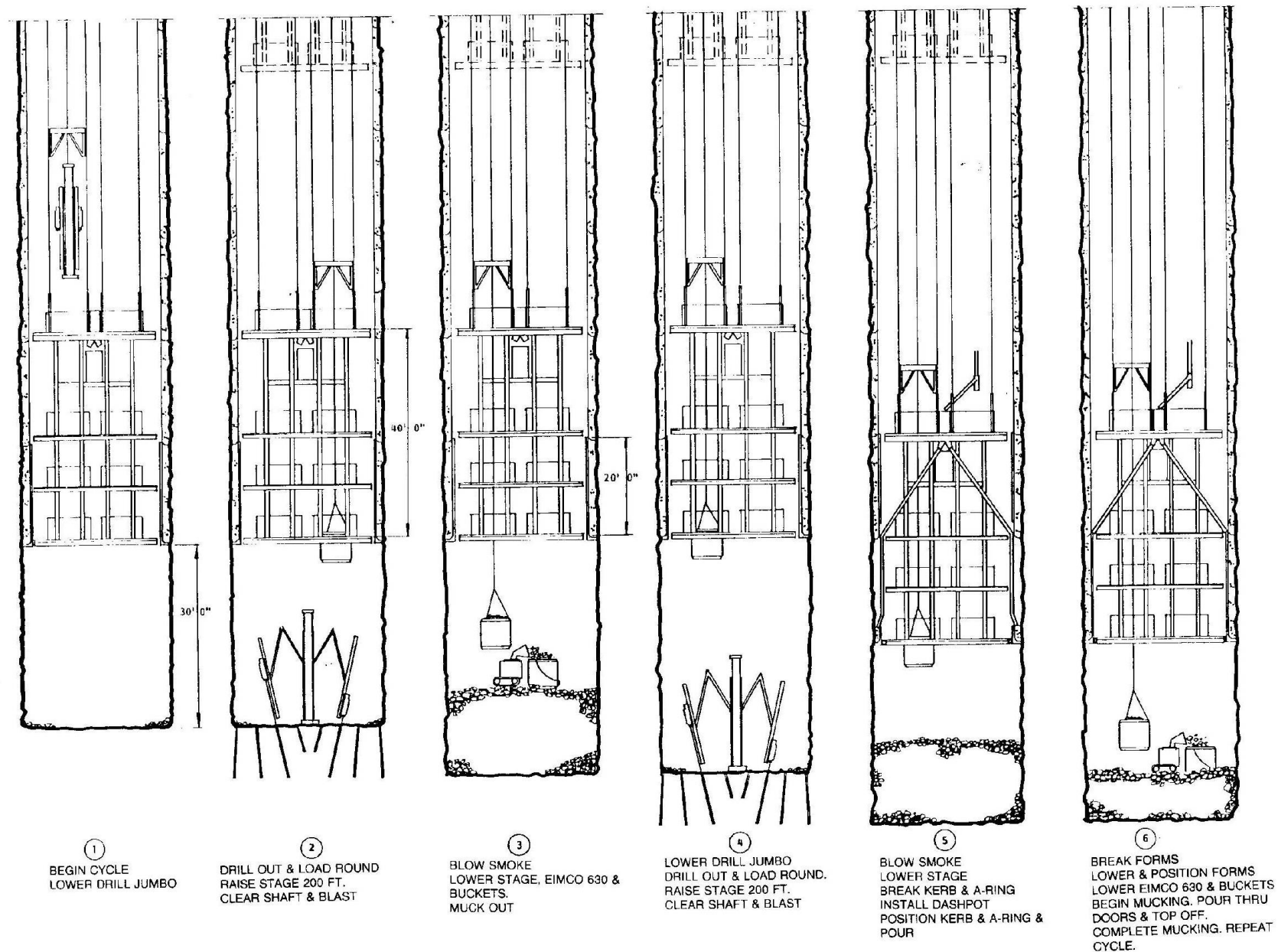
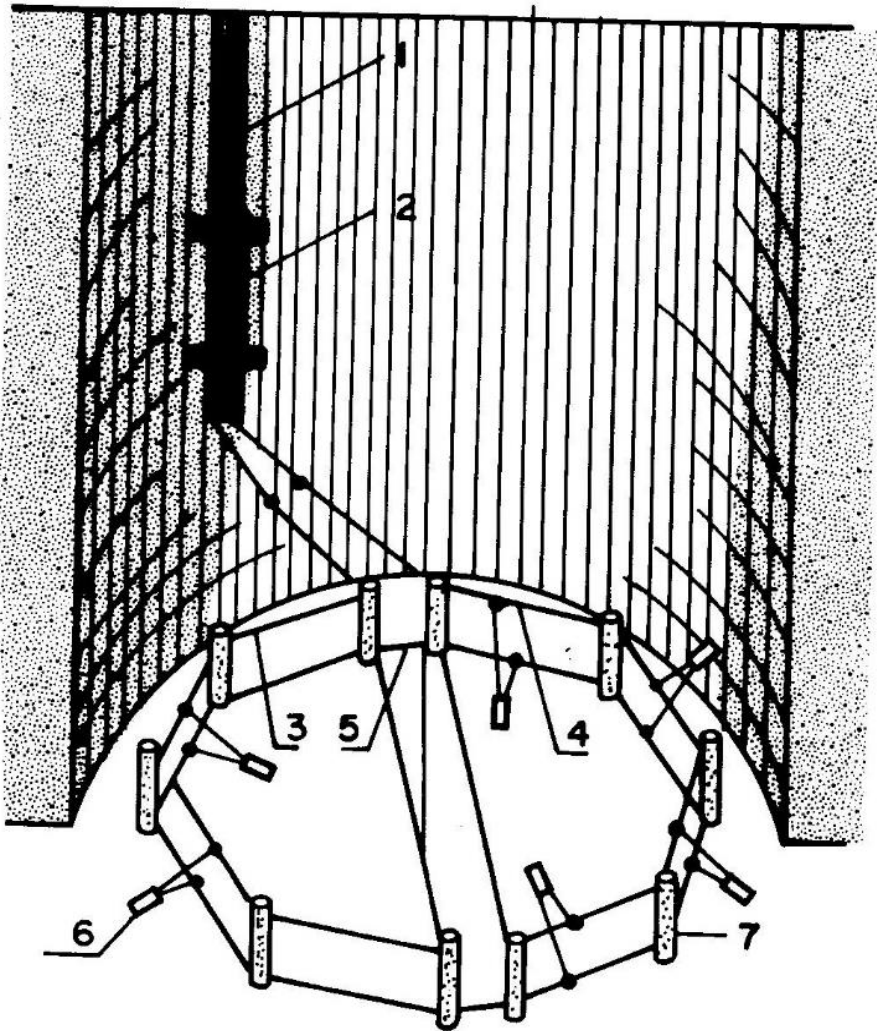


Fig. 17.4.29. Schematic of sinking operations. (After Hynd, 1982.) Conversion factor: 1 ft = 0.3048 m.

Blasting Round I



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Fig. 17.4.28. Three-ring connection of electric caps.

- 1. shot cable
- 2. protecting pipe
- 3-4. active antenna
- 5. passive antenna
- 6. electric caps (in holes)
- 7. wooden stick supporting antennas.

Shaft Drilling Jumbo I

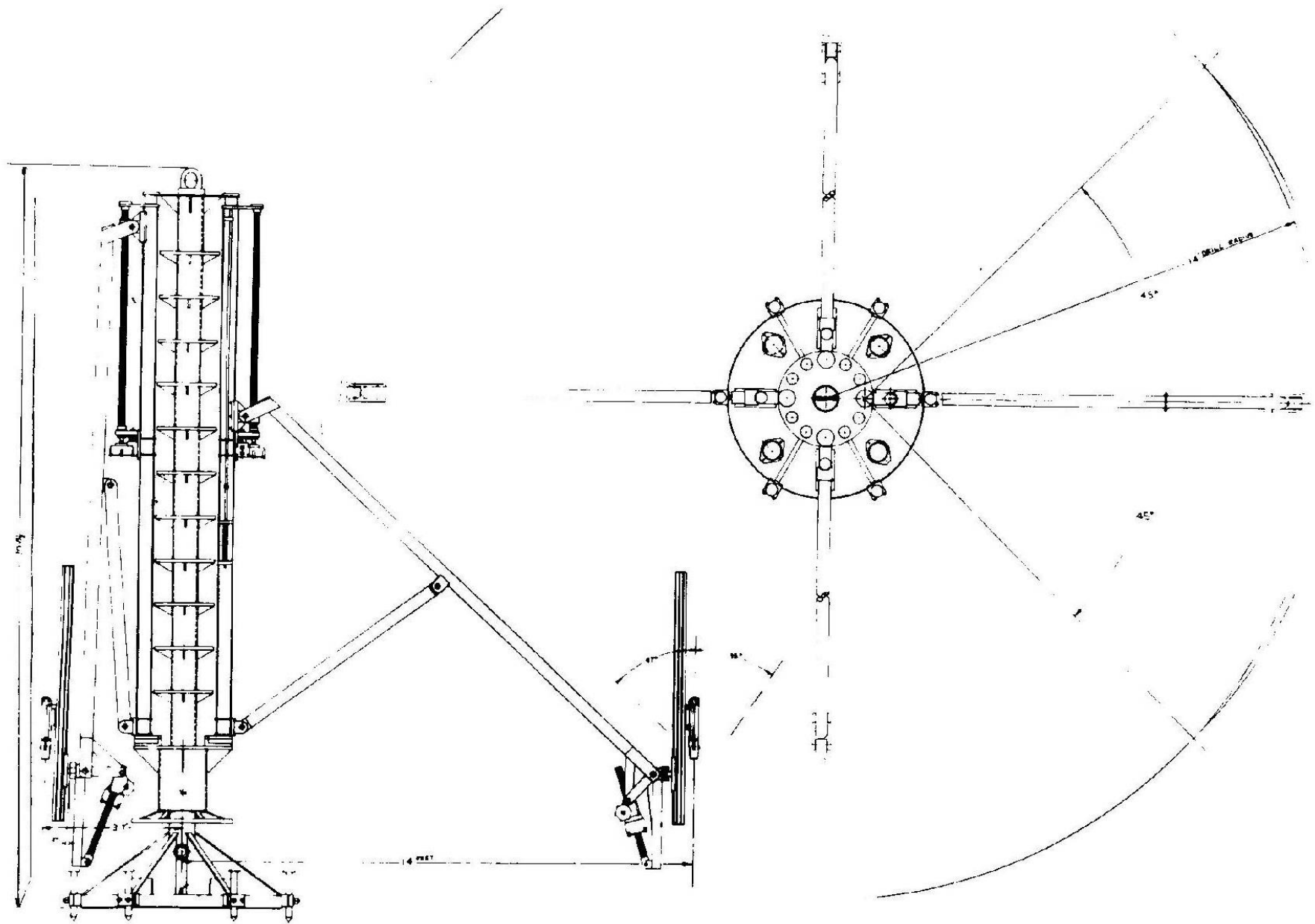
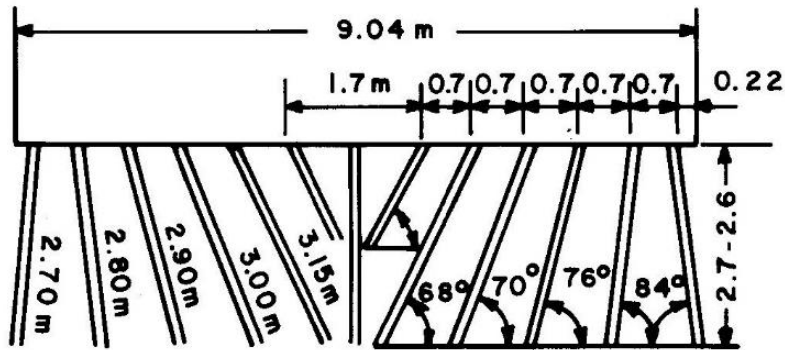


Fig. 17.4.27. ASJB-3B Shaft Jumbo (courtesy: Acme Machinery Co.). Conversion factor: 1 ft = 0.3045 m.

Drilling Round Circular Shaft



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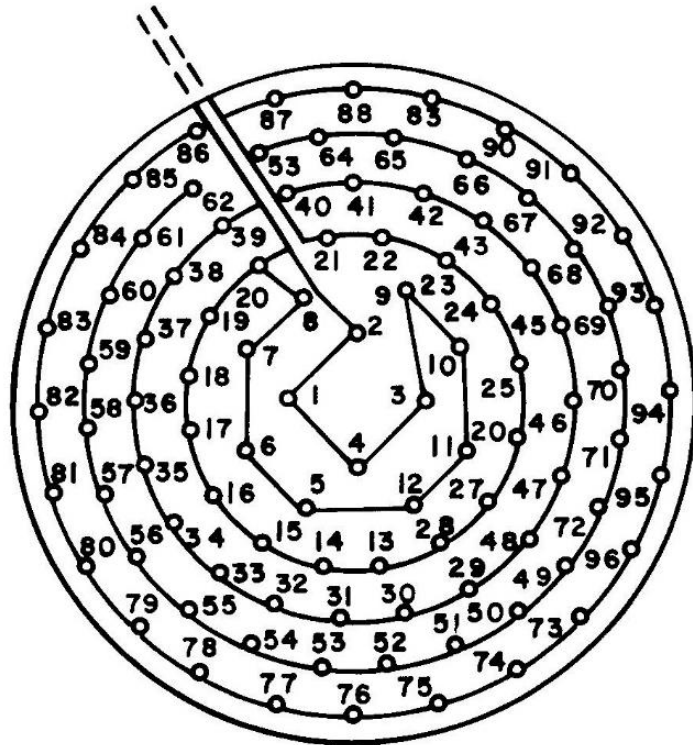
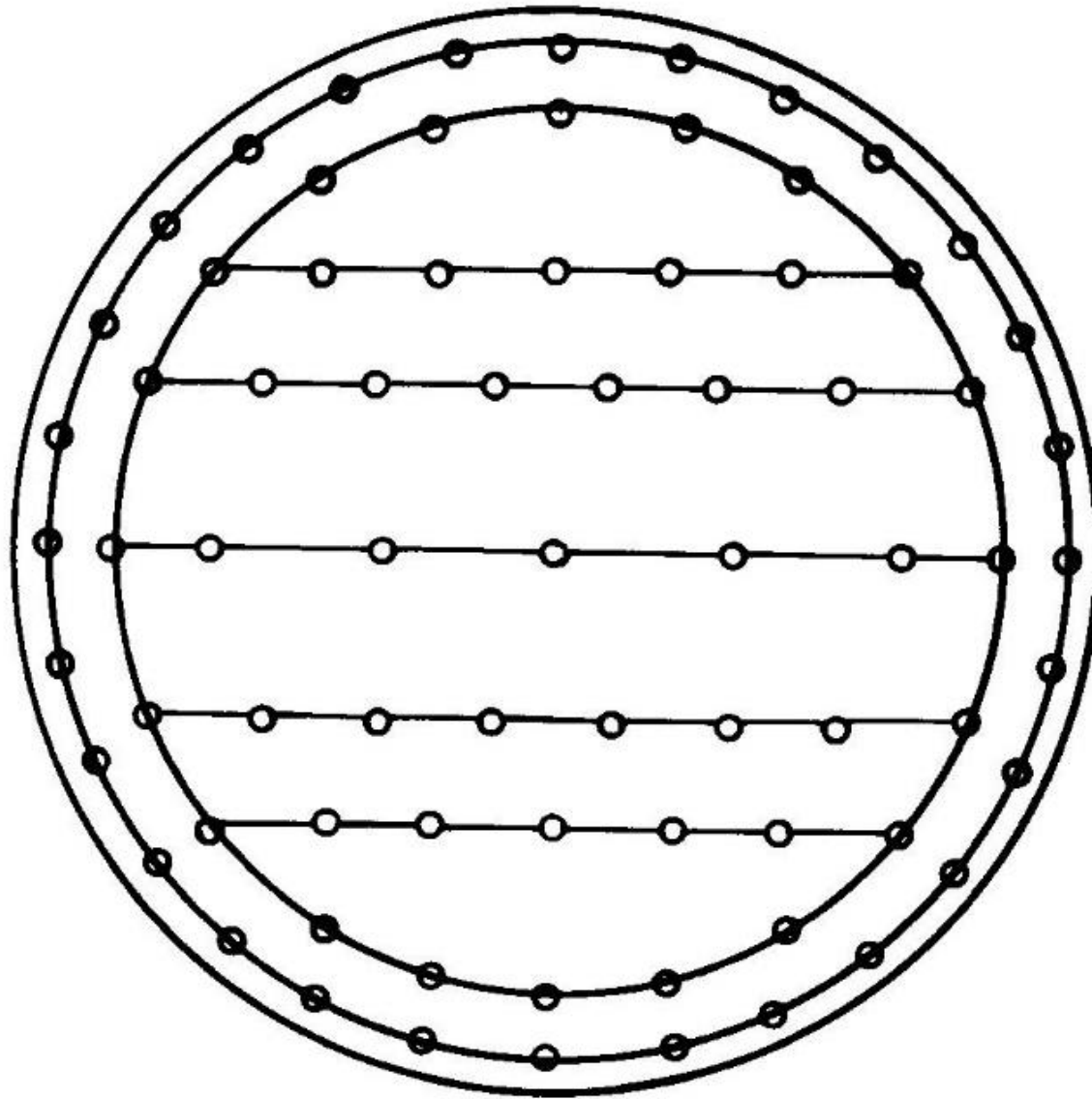


Fig. 17.4.26. Drilling pattern for shaft 27.5 ft (9.04 m) in diameter.
Conversion factor: 1 ft = 0.3048 m.

Drilling Pattern I



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Fig. 17.4.24. Example of drillhole distribution in shaft sunk in inclined strata.

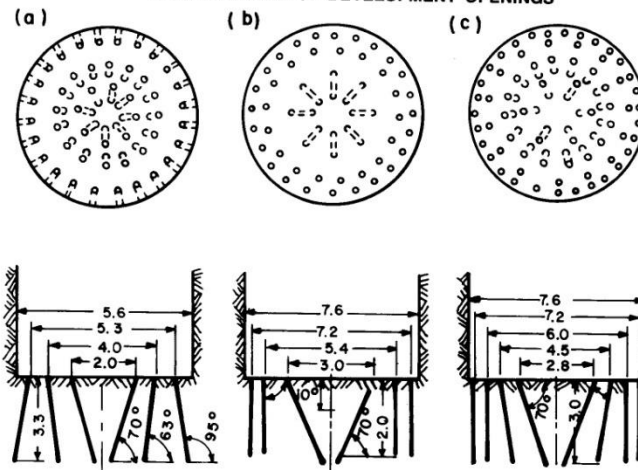


Fig. 17.4.23. Examples of drillhole pattern in circular shaft.
a) in siltstone
b) in shale
c) in sandstone

Dimensions in m. Conversion factor: 1 ft = 0.3048 m.

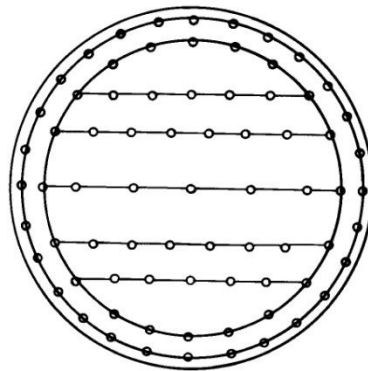


Fig. 17.4.24. Example of drillhole distribution in shaft sunk in inclined strata.

event the hazard of stray current, the ends of wires are short circuited until they are connected to the shot cable.

A clay-sand mixture is used as stemming. In wet shafts, adding to a 10-in. (250-mm) depth substitutes for stemming.

Loading time for one hole is about 6 to 8 minutes and for the whole round, about 60 to 90 minutes. Caps of detonating cartridges can be connected in series, in parallel, or in a mixed manner. Connection in series is the simplest method but also the least reliable. Failure of one cap means a misfire of all the holes.

A mixed connection is done in such a way that a group of holes is connected in series, and particular groups are connected parallel to the main shot cable. An example of a mixed connection is shown in Fig. 17.4.28.

When connections are completed, face lights are pulled up on the stage. The shot master is the last to leave the face. Firing is done by a shot lighter from the surface.

SHAFT VENTILATION. Shaft ventilation is carried out by fan and vent tubing, as described in Chapters 11.6 and 11.7.

MUCKING IN SHAFTS. Mucking in shafts is the most time- and effort-consuming operation, accounting for as much as 50 to 60% of cycle time. Table 17.4.7, gives different factors influencing mucking efficiency. Fig. 17.4.29 shows a schematic of sinking phases with an Eimco 630 loader. Fig. 17.4.30 illustrates the section through a shaft sunk with a cactus grab loading unit.

Mucking efficiency is dependent also on the degree of rock fragmentation and the height of the bucket. For mechanical mucking, medium-sized fragmentation, approximately 5 in. (125 mm), is most efficient.

Generally, in a muck pile after a shot, 70 to 80% of the material is fine fragmentation, and the rest is composed of larger fragments.

Muckers.—Several types of muckers are in use depending on shaft shape, size, depth, and location, as listed below:

1. Scrapers, used in rectangular shafts and very large-diameter shafts (underground bunkers).

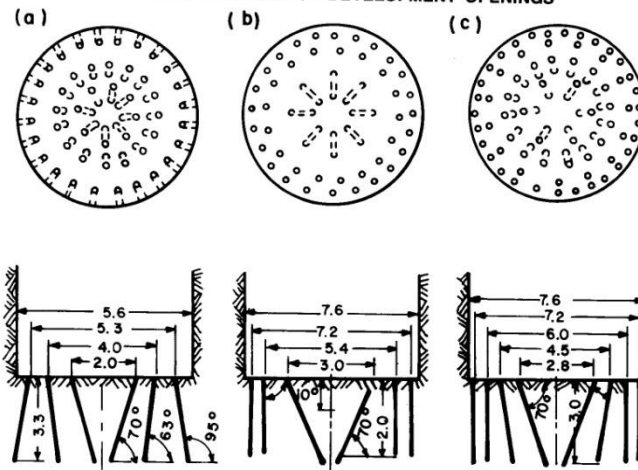


Fig. 17.4.23. Examples of drillhole pattern in circular shaft.
a) in siltstone
b) in shale
c) in sandstone
Dimensions in m. Conversion factor: 1 ft = 0.3048 m.

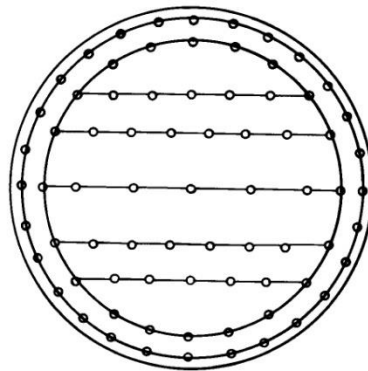


Fig. 17.4.24. Example of drillhole distribution in shaft sunk in inclined strata.

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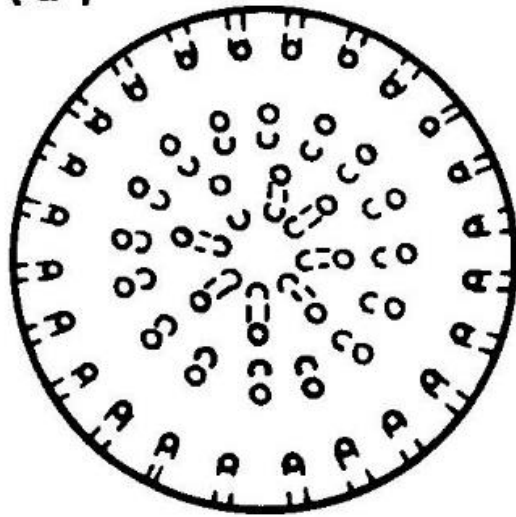
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1. Scrapers, used in rectangular shafts and very large-diameter shafts (underground bunkers).

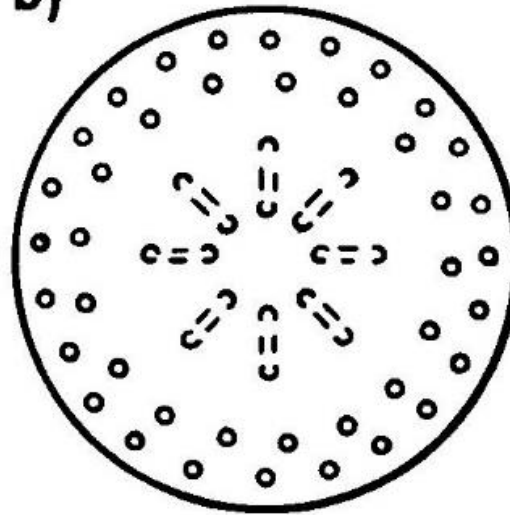
Drilling Pattern in Circular Shaft

(a)



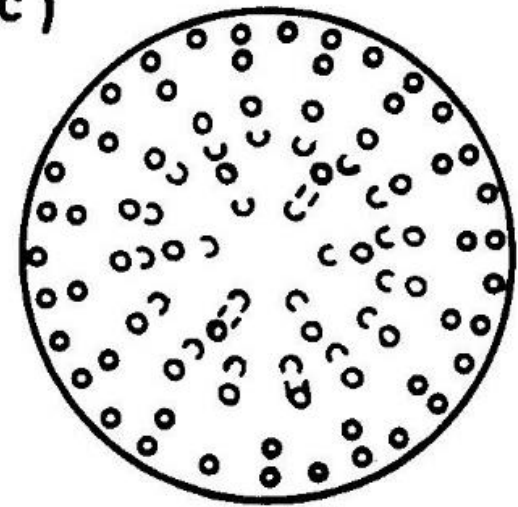
Siltstone

(b)

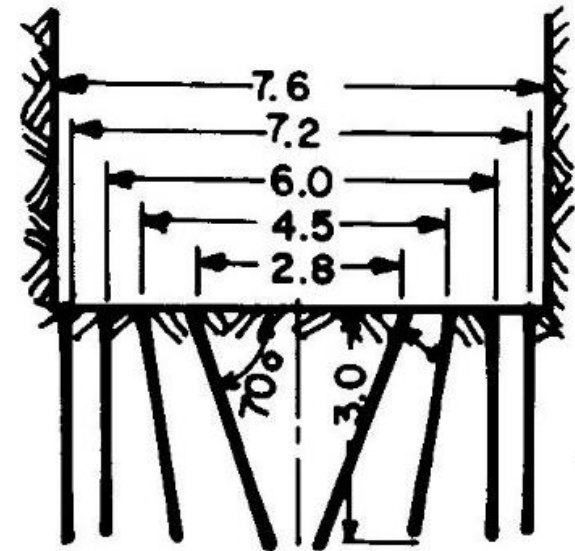
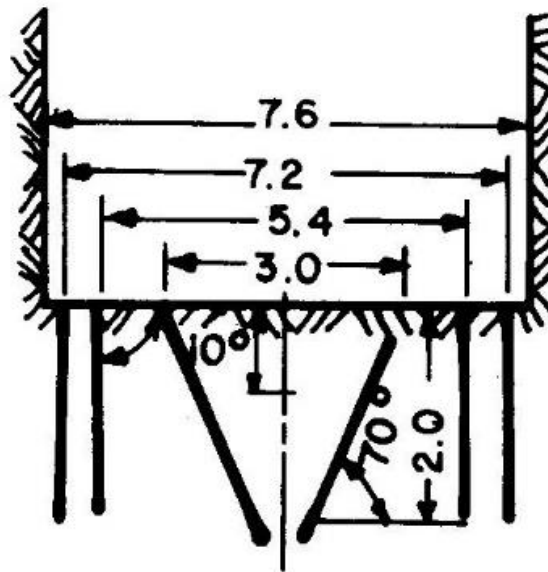
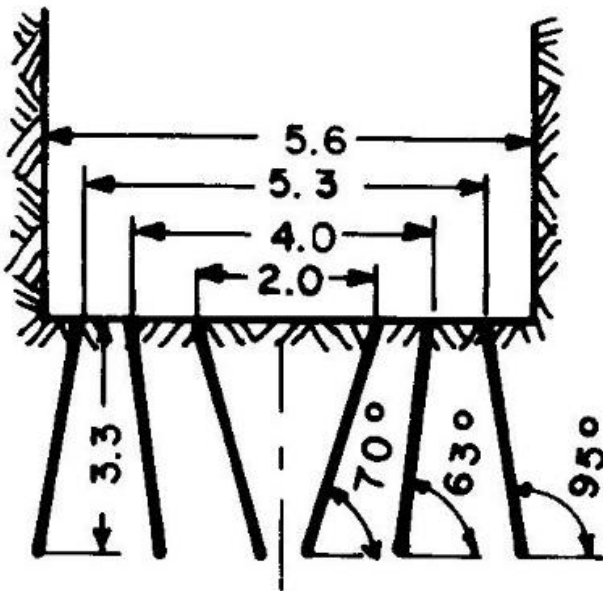


Shale

(c)

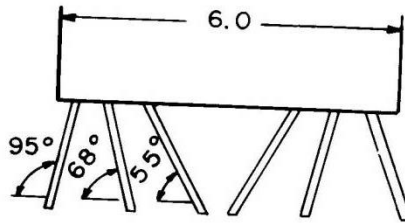
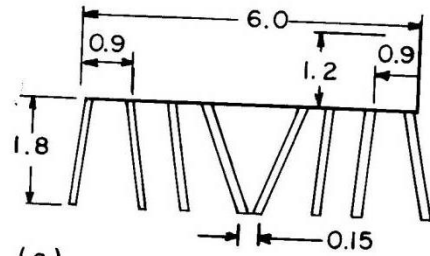
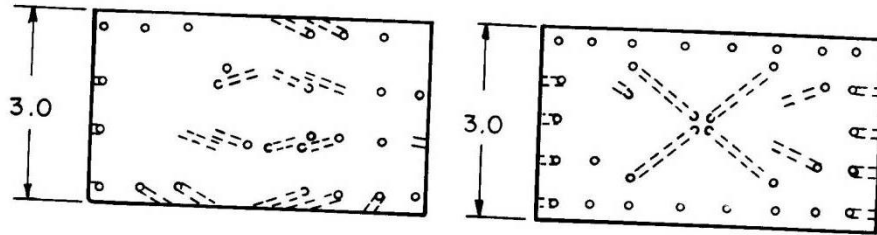


Sandstone



Drilling Round I

*Drilling round in rectangular shaft.

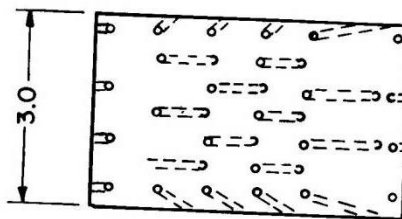
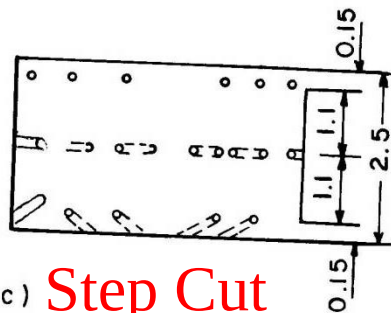
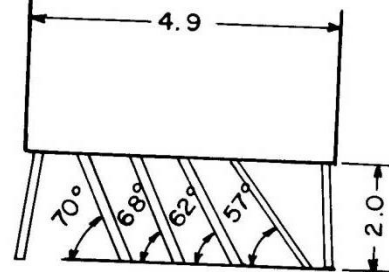
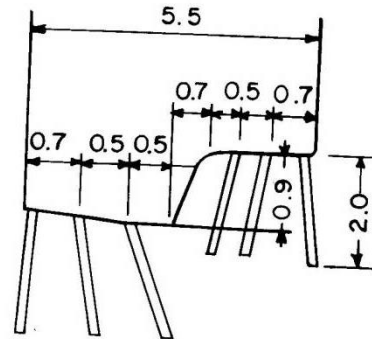


(a)

(b)

Wedge Cut

Pyramid Cut



(c)

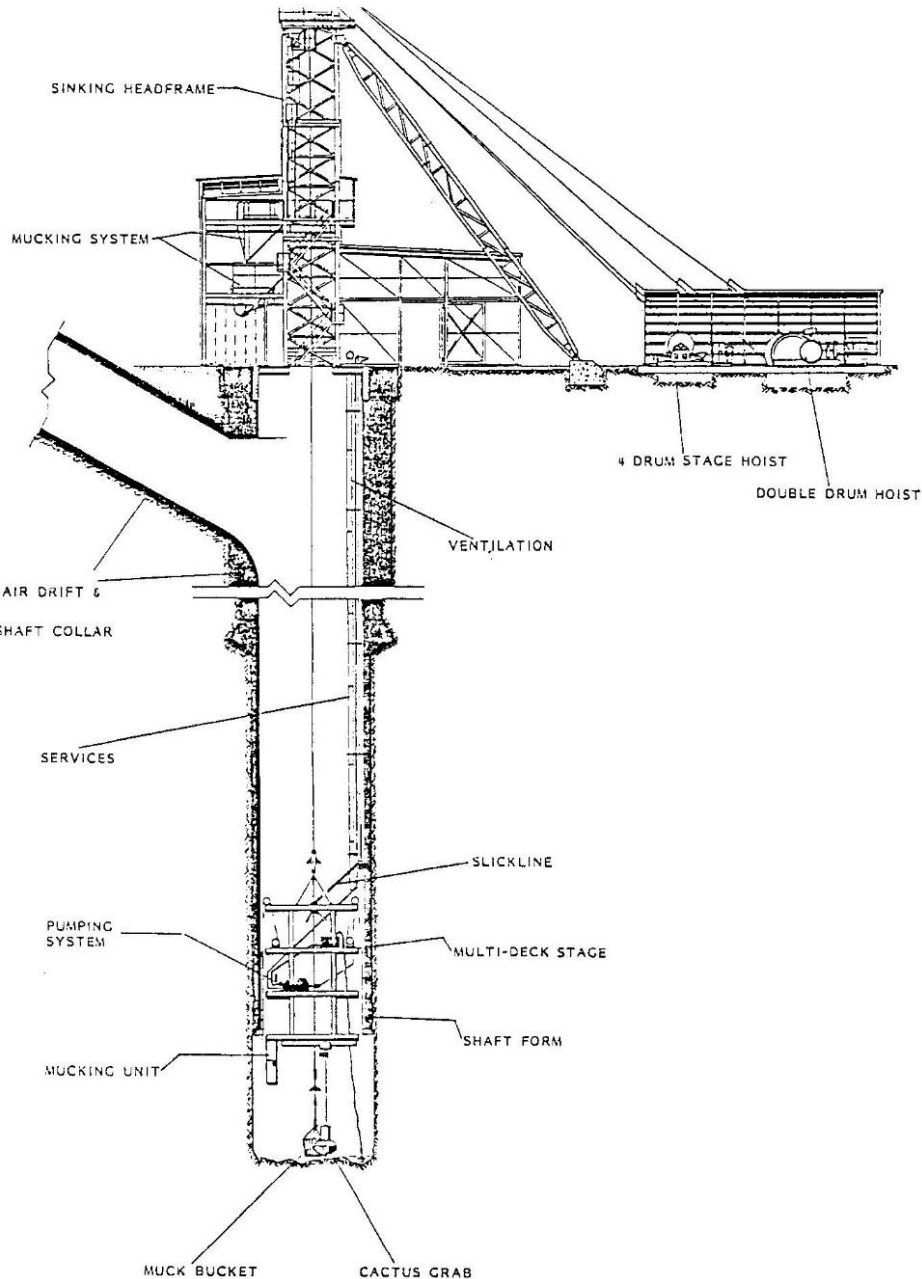
Step Cut

Wedge Side Cut

Drilling Round I

CONSTRUCTION OF DEVELOPMENT OF

*Shaft Sinking Arrangement



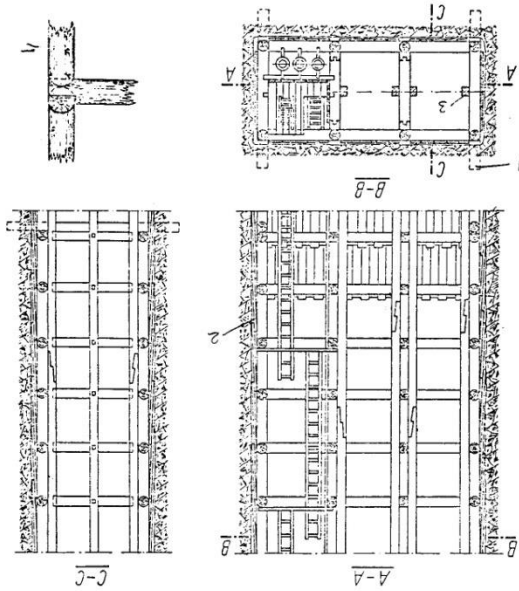


Fig. 17.4.33. Timber lining on posts.
1. bearing set
2. lining with 1-in. (25-mm) boards
3. guides.

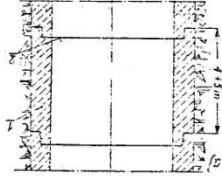


Fig. 17.4.35. Section through the shaft with concrete lining, made with application jump form.
1. concrete
2. seam
Conversion factor: 1 ft = 0.3048 m.

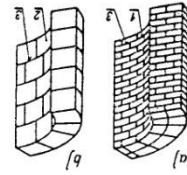


Fig. 17.4.34. Schematic of the cut-out of:
a) brick and b) concrete block
1. brick
2. wedge concrete shaft
3. seams filled with mortar.

where d is permanent lining thickness, b is width of guide rings, c is thickness of driven piles, and f is thickness of guide beams.

$$D_g = D + 2(d + 2b + c + f) \quad (17.4.17)$$

to the following:

D_g (Figs. 17.4.36 and 17.4.37) should be established according

WOODEN PILE WALL. The initial diameter of the shaft collar
currently used methods of placing a concrete pile wall are piles
driven with an outer steel tube with a diameter of 20 in. (520
mm), and drilled piles, with or without slurry, leaving diameters
20 to 48 in. (520 to 1200 mm). The outer steel pipe is forced into
the ground by a pile driver, whereas drilled piles are made by
means of a drilling rig. Where the ground does not assure stabi-

SHEET PILE WALL (STEEL). Construction of a steel sheet
pile wall is shown in Fig. 17.4.38.

CONCRETE (REINFORCED CONCRETE PILE WALL). Two

Open Manual Raise Applicability I

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Open Manual Raise Process I

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Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process II

Open Manual Raise Process III

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Open Manual Raise Process IV

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Open Manual Raise Process V

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Open Manual Raise Survey I

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Open Manual Raise Survey II

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Open Manual Raise Ventilation I

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Open Manual Raise Blasting I

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Open Manual Raise Blasting I

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Open Manual Raise Dressing I

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Open Manual Raise Dressing II

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Open Manual Raise Advantage I

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Open Manual Raise Disadvantage I

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Reference I

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Reference I

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